Finite Element Simulation of Wire Arc Additive Manufacturing: Prediction of Temperature and Stress Evolution During Processing

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ABSTRACT

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This thesis is a research on the Wire Arc Additive Manufacturing (WAAM) process using finite element simulation. WAAM is a metal additive manufacturing process that is based on the deposition of layer-by-layer components, where a metal wire is melted into a stream of an electric arc. It has a high deposition rate and low cost of operation which has made WAAM gain more application in large metallic structures fabrication. Nevertheless, non-uniform heating of the deposition causes too many thermal gradients, large residual stresses, and geometrical distortion, which undermine dimensional accuracy and structural integrity (Montevecchi et al., 2016; Tatemoto et al., 2023).

The current paper is devoted to the numerical modeling of temperature change and stress buildup when the 1 bead of S235 structural steel is deposited. Heat flow was simulated using a transient heat model that used the double-ellipsoidal heat source of Goldak and coupled thermo-mechanical analysis was used to approximate the formation of stress. Using the values of the process parameters determined; the heat input of 40 W/mm, travel speed of 5 mm/s and the inter-pass cooling time of 20 s, the maximum temperature during deposition was about 1500-1600 °C, which was concentrated at the center of the molten pool. The generated thermal gradients created tensile residual stresses of the order of 200-300 Mpa, which also correlate with earlier values in the literature of WAAM research (Bauer et al., 2021; Tatemoto et al., 2023).

The simulations also affirm that process parameters have a strong effect on the quality of parts: heat input is lower or cooling times are increased, the effect on thermal gradients minimized, and consequent material in the form of residual stress and distortion is minimized. All in all, the FEM simulation proves to be highly proficient in simulating the thermo-mechanical nature of WAAM, which makes it a credible predictive instrument of optimizing the parameters and enhancing the quality of metal components made through the WAAM method.

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SYMBOLS AND ABBREVIATIONS

Roman characters

c_p	Specific heat capacity	J/kg·K
I	Welding current	A
k	Thermal conductivity	W/ m·K
p	Pressure	Pa
Q	Total heat input	W
q(x,y,z)	Volumetric heat flux distribution	W/m³
T	temperature	ºC
T_0	Reference temperature	°C
t	Time	s
u	Displacement	m
v	Welding/travel speed	m/s
V	Voltage	V
x(t)	Position of moving heat source	m
ρ	Density	kg/m³
σ	Stress	MPa
σ_{res}	Residual stress	MPa

Greek characters

α	Coefficient of thermal expansion	1/K
η	Process (arc) efficiency	
ε_{elstic}	Elastic strain	
$\varepsilon_{plastic}$	Plastic strain	

$\varepsilon_{thermal}$	Thermal strain
ε_{total}	Total strain

Subscripts

f	Front ellipsoidal zone
r	Rear ellipsoidal zone
res	Residual
0	Reference condition
t	Time dependent variable

Abbreviations

CAD	Computational Aided Dynamics
CFD	Computational Fluid Dynamics
FEA	Finite Element Analysis
FEM	Finite Element Method
GMAW	Gas Metal Arc Welding
GTAW	Gas Tungsten Arc Welding
HAZ	Heat-Affected Zone
DIC	Digital Image Correlation

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1 Introduction

Wire Arc Additive Manufacturing (WAAM) is a metal additive manufacturing process that creates components layer by layer by melting a metal wire with an electric arc. It combines traditional welding techniques with modern digital manufacturing, allowing the production of complex components with high material efficiency and relatively low cost (Pant et al., 2023). Due to its high deposition rate, WAAM is widely used for large structures in several industries, including aerospace, shipbuilding, construction, mining, and energy production (Martina et al., 2012; Williams et al., 2016a). Although many early WAAM studies focused on titanium and aluminium alloys, recent work shows that steels are now one of the most important industrial applications of WAAM. Low-carbon steels, stainless steels, and structural steels are used in heavy industries where large parts, high strength, and low cost are required. WAAM is increasingly applied for repair and manufacturing of steel components such as marine structures, pressure vessels, pipelines, construction beams, tooling, offshore components, and mining equipment (Jin et al., 2020; Long et al., 2021). Structural steels like S235 and S355 are especially relevant because they are widely used in engineering applications due to their good weldability and low material price.

However, WAAM of steel also introduces significant challenges. The repeated heating and cooling cycles generate strong thermal gradients, which can cause residual stresses, distortion, and microstructural changes that influence strength and durability (Chiumenti et al., 2010). Understanding these effects is important because steel components often operate under heavy loads and strict safety requirements. Finite Element Method (FEM) simulation has become a reliable tool for predicting temperature evolution, heat transfer, stress development, and distortion during the WAAM process (Montevecchi et al., 2016a). These simulations help engineers optimize process parameters without expensive and time-consuming experiments.

This thesis investigates a single-bead WAAM process using S235 structural steel. S235 is widely used in industrial applications and provides a good reference material for studying the thermo-mechanical behavior of steel during WAAM. The goal is to predict the temperature and stress distribution for a heat input of 40 W/mm and an inter-layer cooling time of 20 seconds. The results offer a fundamental understanding of thermal and mechanical

responses in steel WAAM and support the optimization of steel-based multi-layer deposition processes.

1.1 Background

WAAM is an additive manufacturing method of metals, constructing additives in layers with a metal wire by melting it using an electric arc. This technique has several advantages, like its high deposition rate and low cost of production large metal components can be fabricated at an efficient rate(Galán Argumedo et al., 2024). this process is widely applicable due to the fact that it uses time-tested welding techniques in various sectors, including aerospace, automotive, and construction., i.e. Gas Metal Arc Welding (GMAW) and Gas Tungsten Arc Welding (GTAW) (Martina et al., 2012) Nevertheless, in spite of these benefits, WAAM continues to have problems associated with heat input, temperature control, and the formation of residual stresses that may result in distortion and lower the quality of the products.

In the WAAM process, the arc is used to expose the metal wire to very high temperatures to cool it down as soon as possible. Such heat cycles provide different expansions and contractions in the material, which cause residual stress to develop. This stress can result in distortion, breaking, and even the fracture of the part in course of service(Ding et al., 2015a). Researchers have used FEM in order to model thermal and mechanical response in WAAM and study such problems and reduce them. The FEM allows the engineer to consider how temperature and stress vary all over the process, and to experiment with parameters such as heat input, welding speed, and cooling time without experimentation (Chiumenti et al., 2010).

The research paper is about the single-bead WAAM simulation of S235 structural steel, a common and weldable material in the field of engineering. The forecasts will be carried out for temperature fields and stress distribution with the help of thermal and mechanical analysis using FEM. The results should enhance knowledge of the interaction between heat and stress in WAAM and optimize the process to minimize distortion and defects in the final parts.

1.2 Motivation

Wire Arc Additive Manufacturing (WAAM) has gained significant attention in recent years because it can manufacture large metallic parts at a lower cost compared to traditional machining or powder-based additive processes. It is especially useful for producing high-strength and customized components in aerospace, automotive, construction, marine, and heavy-industry applications (Galán Argumedo et al., 2024). The process also reduces material waste and provides high design flexibility through its layer-by-layer deposition strategy.

Although these advantages make WAAM attractive for industrial use, the process still faces important challenges. A key limitation is the strong thermal cycling that occurs during the deposition of each bead, which leads to uneven heating and cooling of the material. This thermal behaviour often causes residual stresses, distortion, and changes in the microstructure of the deposited metal (Ding et al., 2015b). A smoother transition is needed to highlight these issues: as industries begin adopting WAAM for structural steels, the thermal and mechanical problems become even more critical because steel is highly sensitive to repeated heating cycles.

For low-carbon and structural steels such as S235 or S355, WAAM can lead to grain coarsening, hardness variations, heat-affected zone (HAZ) softening, and excessive distortion, especially when large heat inputs are used (Jin et al., 2020; Long et al., 2021). If these effects are not controlled, components may suffer from cracks, dimensional inaccuracy, or reduced mechanical performance during service. To address these challenges, the Finite Element Method (FEM) provides an effective way to study the thermo-mechanical behaviour of WAAM. FEM simulations can predict temperature evolution, stress formation, and deformation without the high cost of experimental trials (Chiumenti et al., 2010). Previous studies have shown that FEM is a reliable tool for analyzing heat flow and stress development in metal additive manufacturing, including both aluminium, titanium, and steel alloys (Galán Argumedo et al., 2024; Gokuldoss et al., 2017).

In this thesis, S235 structural steel is selected for the simulation of a single-bead WAAM process. S235 is widely used in engineering due to its good weldability and industrial relevance. The aim is to create a thermo-mechanical model that examines how process parameters such as heat input and cooling time affect temperature distribution and residual

stress formation. The results of this study are expected to support the optimization of steel WAAM processes, reduce distortion, and improve the final quality of WAAM-manufactured components.

1.3 Objective

This thesis aims to expand the understanding of computer-based simulation for the Wire Arc Additive Manufacturing (WAAM) process. Although WAAM is an effective technique for producing large metal components, it still faces challenges such as overheating, residual stresses, and deformation due to repeated heating and cooling of each deposited bead. To address these issues, reliable simulation tools are needed to predict heat flow and stress evolution throughout the process.

The main purpose of this thesis is to develop a Finite Element Method (FEM) simulation model for a single-bead WAAM process using S235 structural steel. The model will be used to predict temperature distribution and residual stress formation, allowing a deeper understanding of the mechanisms that lead to cracking, bending, and distortion during deposition.

To achieve this purpose, the thesis defines the following specific and measurable objectives:

1. Simulate the thermal field of the WAAM process using Goldak's double-ellipsoidal heat source model to predict temperature variation during bead deposition.
2. Evaluate residual stresses and deformation generated due to thermal cycling by performing a coupled thermo-mechanical FEM analysis.
3. Investigate the influence of key process parameters, including a heat input, travel speed, and an interlayer cooling time, on temperature gradients and stress development.
4. Validate the simulation trends by comparing results with findings reported in earlier studies (Safyari et al., 2024) ensuring accuracy and reliability of the developed model.

5. Practical adjustments to WAAM parameters based on the simulation outcomes in order to reduce stress, control distortion, and improve the overall quality of WAAM-manufactured steel components.

The final outcome of this thesis will be a useful FEM-based simulation model that can help engineers optimize WAAM more effectively and improve the reliability of structural steel parts produced using this technology.

1.4 Research Problem and Research Questions

WAAM is one of the most promising technologies in metal additive manufacturing because of its capability to produce large and customized metallic parts at lesser cost and within a short time compared with other methods. This process has attracted growing interest in industries that include aerospace, construction, and automotive due to its efficiency and adaptability. However, WAAM encompasses complex thermal behavior wherein every single deposited bead is reheating the material underneath. Repetitive heating and cooling result in non-uniform temperature distributions and cause residual stresses and distortion in the finished part.

The problem involves not just visual or dimensional defects, but also mechanical integrity involving the component. High residual stress levels weaken the part, cause cracks, or even lead to failure during operation. These thermal and stress issues have impeded the wider diffusion of WAAM in high-precision and structurally reliable industries (Martina et al., 2012; Nguyen et al., 2023). Hence, there is a need to develop tools that can predict temperature and stress behavior during the process to aid in enhancing quality and reliability.

The finite element method has emerged as one of the most capable tools for understanding such interactions. FEM allows researchers to simulate the temperature field and stress distribution for different process conditions without necessarily having to conduct several physical experiments. Parameters such as heat input, travel speed, and inter-layer cooling time can be changed to study the effects on the thermal and mechanical behavior of the material. (Chiumenti et al., 2010) conducted a study in which many previous studies, including by (Buchanan & Gardner, 2019; Gokuldoss et al., 2017) analyzed either thermal

or mechanical aspects separately, leaving a gap in how both interact simultaneously during WAAM.

The main research problem of this thesis is to determine how Finite Element simulation can be used to predict both temperature variation and residual stress formation during a single-bead WAAM process using S235 structural steel, and how these predictions can help improve the quality and reliability of the final part. Based on this problem, the study addresses the following research questions:

1. How does the applied heat input influence temperature distribution, thermal gradients, and resulting residual stresses during the WAAM process?
2. How do travel speed and inter-layer cooling time affect the thermal cycles, cooling rates, and stress formation in S235 structural steel?
3. How accurately can a transient thermal model and a coupled mechanical model predict temperature evolution and residual stresses in a single-bead WAAM simulation?
4. To what extent can the simulation results help minimize distortion and improve the process control of WAAM for structural steel components?
5. How do the predicted simulation trends compare with previously published experimental and numerical studies?

The outcomes of this thesis will produce a validated FEM-based model that strengthens understanding of the WAAM process. The results will provide a more accurate and affordable method for predicting temperature and stress, optimizing process parameters, and improving the quality and consistency of steel components manufactured by WAAM.

1.5 Research Scope and Research gap

This thesis focuses on the numerical simulation of the WAAM process using the Finite Element Method (FEM). Its objective is to predict the evolution of temperature and residual stress when a single track of S235 structural steel is deposited using an electric arc as the heat source. Only thermal and mechanical simulations are considered in this work. While no experimental validation was conducted, the results obtained will be compared with findings

from established studies to ensure reliability and accuracy (Galán Argumedo et al., 2024), (Gokuldoss et al., 2017) This model developed in the present research would aim at providing insight into the distribution and flow of heat as well as residual stresses and thus would allow the practitioner to understand how modifying process parameters affects the outcome in WAAM.

In this work, a single-bead track geometry has been considered instead of multi-layer or complex geometries to maintain the simulations simple and facilitate clear interpretation of the results. This allows studying the main physical phenomena-heat transfer, temperature evolution, and formation of stresses-isolated from the unnecessary computational complications (Chiumenti et al., 2010). In the simulation, primary attention is given to the influence of heat input, travel speed, and inter-layer cooling time on thermal and stress fields. Phase transformation, grain refinement, and microstructural evolution parameters are outside the scope of the present study, since for that, more sophisticated multiphysics modeling and experimental data would be required, as highlighted by (Ding et al., 2015b; Y. Zhang et al., n.d.).

Despite many studies on WAAM, significant gaps remain in understanding the coupled thermal and mechanical behaviour of the process, particularly through simplified and practical numerical models. Much of the literature examines thermal fields or stress development separately, and many existing models are applied to complex geometries that complicate validation (Martina et al., 2012) Furthermore, while WAAM simulations of titanium and aluminum alloys are well documented, studies focusing on structural steels remain limited (Hadjipantelis, 2025; Williams et al., 2016b). This creates a shortage of validated thermo-mechanical WAAM models specifically for steels, even though low-carbon and structural steels are widely used in industrial manufacturing, construction, and repair applications (Sampaio et al., 2023; Zhao et al., 2011).

This research attempts to address this gap by developing a simplified yet realistic FEM-based thermo-mechanical model for the WAAM process using S235 structural steel. The model is designed to simulate temperature and stress formation under practical conditions and to provide reliable predictions for a single-bead deposition (Zhao et al., 2011). The outcomes are expected to improve understanding of thermal and mechanical responses during WAAM of steel, thereby contributing to reduced residual stress and distortion in

industrial applications. Ultimately, the thesis supports FEM as a valuable tool for process design, parameter optimization, and quality improvement in metal additive manufacturing.

2 Literature Review

The literature on Wire Arc Additive Manufacturing (WAAM) highlights the strong coupling between process parameters, heat-source behaviour, material properties, and the thermo-mechanical responses that arise during deposition. Understanding these interactions is essential for accurate simulation and process optimisation. As illustrated in finite-element modelling of WAAM requires defining geometry, heat source, boundary conditions, material data, and numerical parameters, which together enable predictions of temperature fields, displacements, strains, thermal cycles, distortions, and residual stresses. This section reviews the key findings from prior research on WAAM process fundamentals, heat-source modelling, thermo-mechanical behaviour, and simulation practices, providing a foundation for the modelling approach used in this thesis.

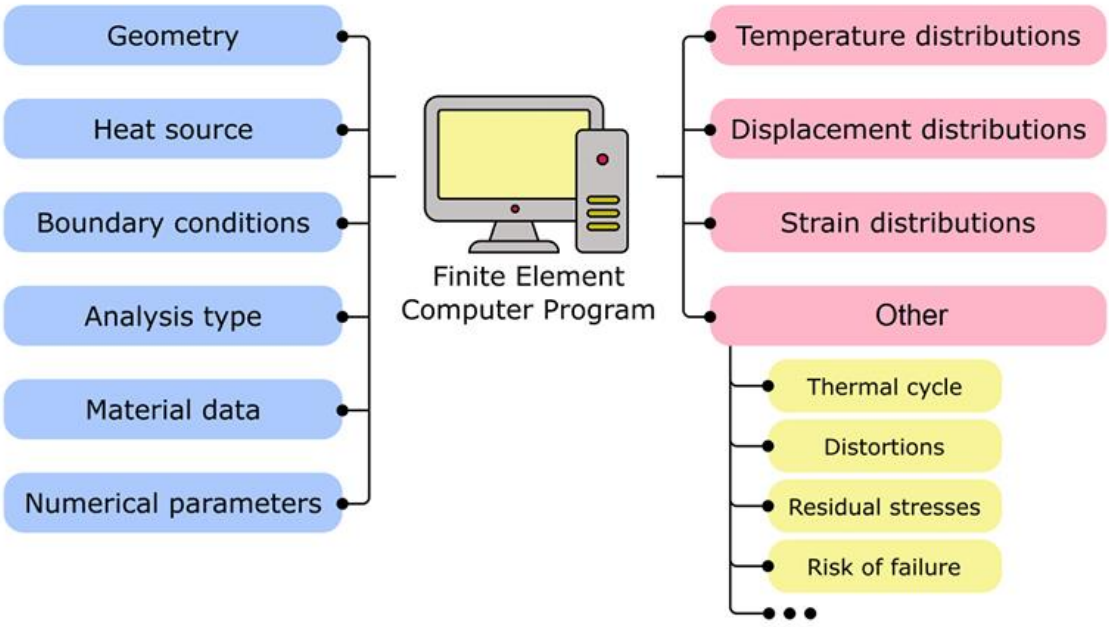


Figure 1 Finite element modelling environment in the macro-scale simulation of WAAM (Sampaio et al., 2023)

2.1 Fundamentals of Wire Arc Additive Manufacturing (WAAM)

A directed-energy deposition method designated as Wire Arc Additive Manufacturing (WAAM) is a method that is used to melt a metallic wire using an electric arc and layer-by-layer deposition of the molten metal to create near-net-shape products. Due to the fact that the process is based on a well-developed arc-welding technologies, WAAM provides a number of benefits including high deposition rates, low amounts of material waste, and the possibility to cast large metallic objects based on the CAD models (Williams et al., 2016b); (Ding et al., 2015b) These features render WAAM a desirable alternative to industrial applications in the aerospace, maritime, energy, and construction industries, where giant, powerful, and individual metal parts are needed.

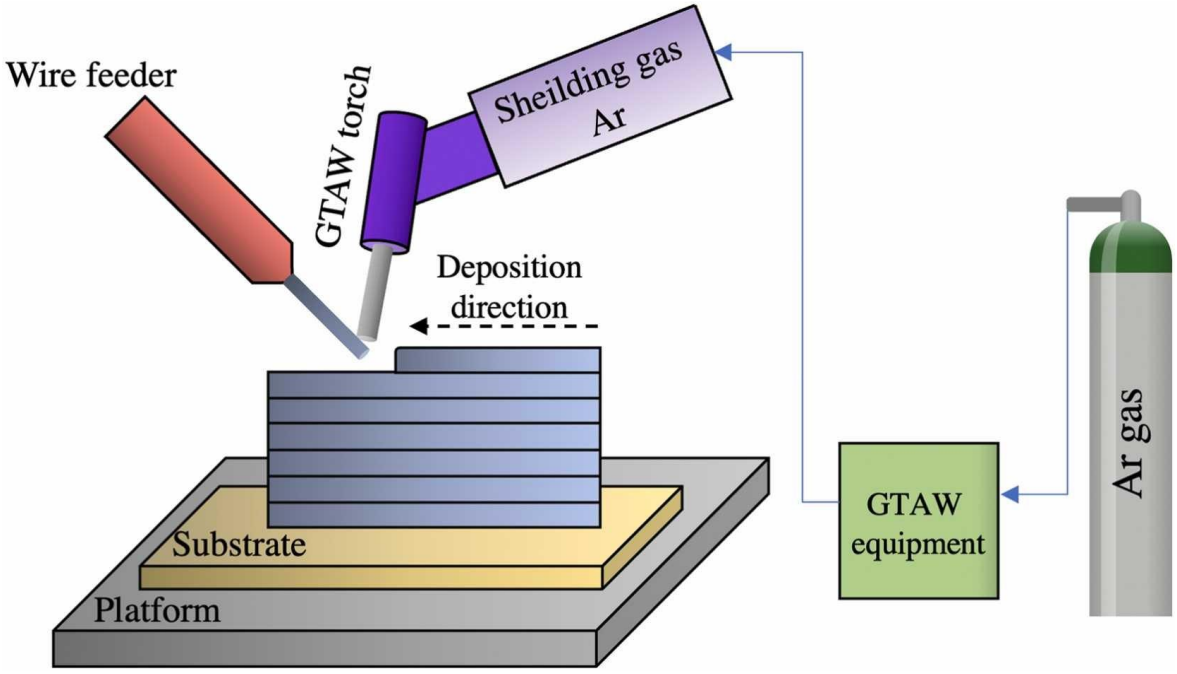


Figure 2 Schematic diagram of wire arc additive manufacturing (Safyari et al., 2024)

Although it is cost-efficient and efficient, WAAM presents difficulties because of its high thermal cycling. In deposition, large heat pulses are repeatedly put on the workpiece; this creates steep temperature gradients that affect the cooling behaviour, the shape of the melt pool, solidification rate, mechanical properties, and internal stress formation (Frazier, 2014). And it in such combination is essential to high-dimensional accuracy, structural integrity, and able part performance. The numerical simulation particularly by the Finite Element Method (FEM) has since emerged as one of the tools in forecasting the thermal field, stress development, and the general behaviour of a process (Montevecchi et al., 2016b); (Wang et

al., 2022) An overview of the major WAAM principles that will apply in this thesis is below but is not exhaustive. .

2.1.1 Principles of WAAM

WAAM involves arc-welding, e.g., Gas Metal Arc Welding (GMAW), Gas Tungsten Arc Welding (GTAW) and Plasma Arc Welding (PAW). Both of the methods use continuous wire feedstock that is melted and deposited on a substrate. GMAW is most popular in the WAAM because it has a high deposition rate, is inexpensive and most alloys work well with it, though GTAW and PAW have more stable arcs and are more controllable at the cost of a slower deposition rate (Martina et al., 2012).

Unlike the Powder Bed Fusion (PBF) which is a powder-based technique of additive manufacturing, WAAM uses no metallic powder, removing the dangers of handling the particles, lowering the cost of materials and increasing the sustainability of production. It is also noted that WAAM has significantly higher deposition rates (as many as 10x higher), which is why it is especially useful in large-scale components that would be both inconvenient and economically unfeasible to produce using powder-based systems (Derekar, 2018). The feedstock wire also allows good utilisation of material, with the minimum amount of waste produced.

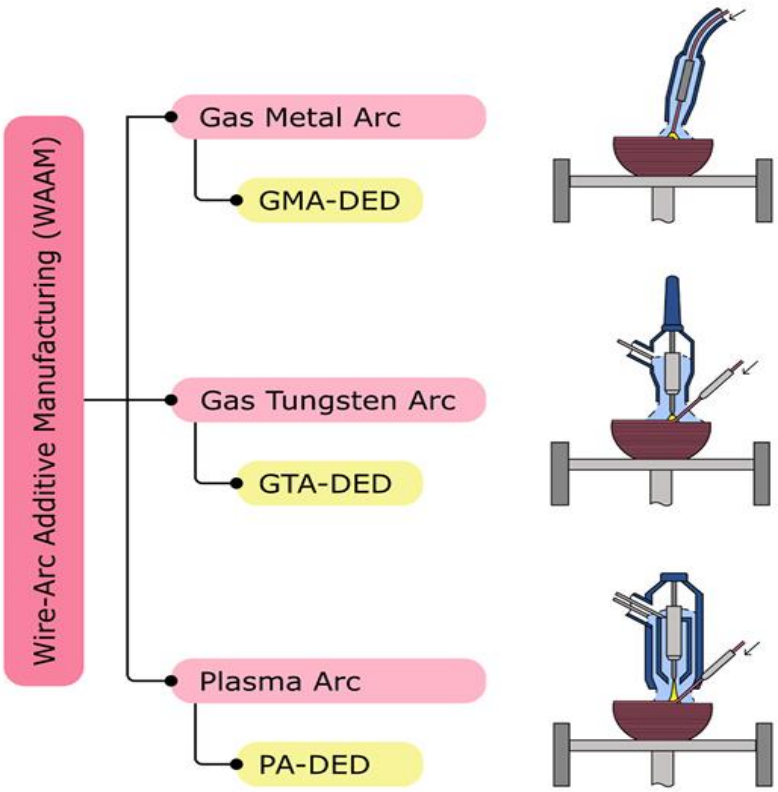


Figure 3 Classification of wire-arc additive manufacturing (WAAM) processes (Sampaio et al., 2023)

Nevertheless, high heat input and large melt pools, which are the same benefits of WAAM, present some challenges to the process. These result in major thermal identities since the arc heats the material in a continuous manner whereas other parts of it cool down at a fast rate. These gradients affect the shape of the melt pool, the kinetics of the phase transformation, grain structure, and accumulated residual stress (Kokare et al., 2024) In the absence of control, such effects may cause distortion, warping, absence of fusion, porosity or cracking. Therefore, accurate heat transfer and solidification behaviour control is required in the determination of the mechanical behaviour of the final component.

2.1.2 Process Parameters and Deposition Strategies

WAAM highly depends on process parameters to determine the quality of parts. The major parameters are arc voltage, welding current, wire feed speed, and travel speed, shielding gas flow, interlayer cooling time, as well as motion of the torch. A combination of these factors predetermines the amount of heat input, which regulates the size of the melt pool, depth of penetration, rate of solidification, and development of microstructural elements (Ding et al.,

2015b) An example would be that high travel speeds minimize the amount of heat input and can lead to inadequate melting of the wire or insufficient bonding whereas low travel speeds would lead to heat retention and excessive distortion or roughness of the surface.

Besides parameter optimisation, deposition strategy is also a very crucial aspect of controlling the heat distribution and distortion. Raster paths, zig-zag paths, oscillation, and spiral paths are some methods that can determine the amount of thermal accumulation and residual stress distribution (Baier et al., 2022) The control systems are very advanced CAM or robotics and are usually implemented to achieve uniform bead geometry and controlled heat flow. The concentration of stresses can be greatly minimised, geometrical accuracy can be enhanced, and the mechanical behaviour of the printed component can be stabilised with the help of optimised strategies (Williams et al., 2016b)

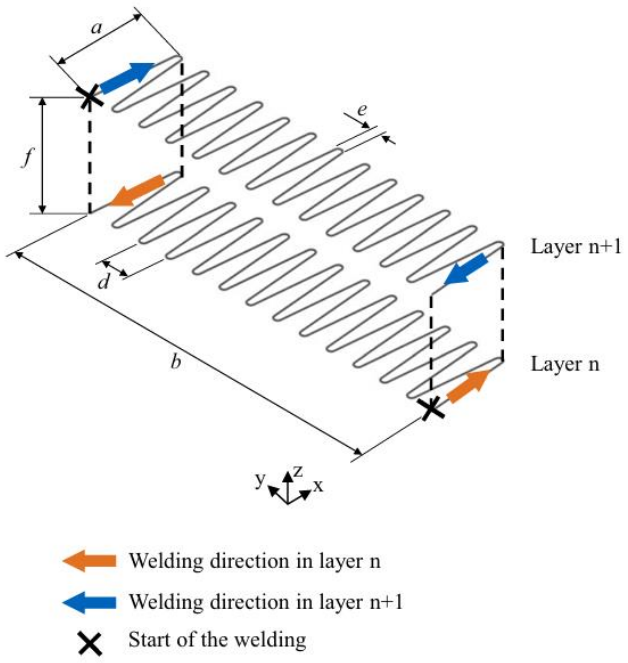


Figure 4 Build-up strategy (Baier et al., 2022)

The key of a successful WAAM is to balance the parameter set and choose a proper course of deposition that ensure uniform heat flow. With these control, there will be better bead consistency, lower distortion, and mechanical properties of the final component. On the other hand, the lack of proper parameter control is likely to lead to layers misalignment, internal flaws, porosity, and erratic microstructure (Gurmesa & Lemu, 2023).

2.1.3 Applications and Challenges

The capability of printing large and personalized metallic parts at a lower cost and within shorter periods has seen WAAM integrated in some industrial applications. WAAM is used to create ribs, brackets, wing parts, and titanium structural parts that save a large percentage of the material used over subtractive technology in aerospace (Martina et al., 2012); (Williams et al., 2016b) WAAM has found application in maritime and offshore engineering to repair or manufacture propellers, stiffeners and large steel articles. In the same regard, under energy and construction, WAAM is used in the manufacture of pressure vessels, support structures, and reinforcement elements (El Kadi, 2006).

Nevertheless, even with its potential in industry, WAAM has a serious number of limitations. The most outstanding problem is the development of residual stresses as a result of repeated thermal cycling. Asymmetrical heating and cooling result in limited contraction, which may result in dimensional distortion, warping, and even cracking when stresses are beyond the material strength in the locality (Lindgren, 2006); (Wang et al., 2022) These problems have a direct impact on fatigue performance, structural reliability and long term service behaviour of WAAM parts (Vázquez et al., n.d.).

The other issue is rather coarse surface finish and exposed layer pattern which usually need post-processing like milling or grinding (Williams et al., 2016b) Also, WAAM components are parameter sensitised, i.e. minor differences in arc power, cooling period, or travel speed can cause a major change in thermal behaviour, mechanical properties, and general part integrity (Xiong et al., 2018).

Certain alloys like aluminium provide additional challenges because of their high thermal conductivity, low viscosity and porosity. These types of property demand better controlled heat management and more sophisticated control to have stable deposition(Gupta & Mulik, 2025).

Current studies have concentrated on predictive FEM modelling, enhanced control algorithms, optimisation of deposition strategies and enhanced knowledge of thermo-mechanical behaviour. The focus of these developments is to ensure that WAAM becomes a more viable and dependable technology in the production of industries in the future.

2.2 Thermal-Mechanical Response of the WAAM Process

This is because Thermal and mechanical behaviour in Wire Arc Additive Manufacturing (WAAM) is very important in determining the end quality, accuracy and structural performance of the printed pieces. The fact that WAAM uses an electric arc to generate the heat source means that each deposited bead deposits a lot of heat into the substrate. During the heating process of the material, the rapid peak heating is followed by steep cooling, resulting in complex thermal cycles in the whole of the component as the torch travels through the deposition line. The cycles have a potent impact on the distribution of temperature, solidification behaviour, formation of residual stress, and eventual mechanical properties of the resulting part (Williams et al., 2016b); (Frazier, 2014)

The heating and cooling that happens differently in the deposited layers make the material to expand and contract differently. This discrepancy in thermal contraction is the primary cause of the development of residual stress in WAAM. These effects are aggravated by high heat of input and repetitive reheating of the layers that have already been deposited. The thermal behaviour cannot be controlled properly, which can cause distortion, warping, cracking, or reduced fatigue performance- all of which continue to be some of the most important constraints of WAAM to be used in industry (Wang et al., 2022) Due to this fact, the relationship between the development of the temperature, the creation of stress, and the parameters of the processes is necessary to enhance stability and reliability of the components of WAAM.

These thermal and mechanical conditions are predicted more often on the basis of Finite Element Method (FEM) simulation. Numerical modelling can be used to set up temperature gradient, stress build-up and thermal history in advance of printing. It facilitates the process parameter optimisation and minimises the cost of the experiment and material wastage (Montevecchi et al., 2016b); (Baier et al., 2022)

2.2.1 Heat Flow, Temperature Gradients, and Cooling Dynamics

The thermal conductivity in WAAM is controlled by the arc energy, thermal properties of the material and the deposition route. As the wire is melting under the influence of the electric arc, a molten pool is created and the heat spreads quickly into the substrate around.

Owing to the focussing effect of the heat source, the hottest part of the arc is beneath the developed arc, whereas the far-off regions is relatively cooler. This results in sharp thermal gradients which affect the cooling rate, shape of the melt pool and solidification behaviour (Goldak et al., 1984; Goldak and Akhlaghi, 2005).

The benefit of WAM is not precision but productivity, scalability, as well as geometric versatility. WAAM may place several kilograms of material per hour and can even create very large parts, including curved or freeform ones that are challenging to make using powder-based methods (Williams et al., 2016b) The process also minimizes the amount of waste produced during the process of materials since the feedstock of wire is utilized effective and machining allowances are less than those found in the subtractive processes.

Nonetheless, the amount of heat to be put into deposition to ensure a fast deposition is the cause of various challenges. Unequal temperature concentration can cause the swelling of the molten pool, dilution, and decrease in bead stability. Non-homogeneity of cooling encourages the residual stress and distortion. It has been found that the thermal behaviour is very sensitive to current, voltage, travel speed, as well as interlayer cooling time (Ding et al., 2015b); (Kokare et al., 2024) Better thermal control is thus vital towards acquiring reliable bead morphology and part stable geometry.

Thermal modelling by FEM has proved to be a sure way of understanding such mechanisms. Numbers One of the uses of numerical simulation by researchers is to anticipate peak temperatures, profiles through which the heat transmission influences zone (HAZ) areas, cooling claims and the influence of various process circumstances on thermal zone. The predictive modelling has been found to be helpful in obtaining more homogenous temperature distributions and reducing heat build-ups in long build (Montevecchi et al., 2016b);

2.2.2 Residual Stress Formation and Resulting Distortions

One of the most important problems in WAAM is residual stress formation as it has a direct influence on the dimensional precision and the mechanical performance. As it is deposited the molten metal that has been added heats up and expands. It contracts and freezes, but is

held back in its contraction by neighboring solid material. This results in internal stresses which are retained in the structure once the cooling is finished (Lindgren, 2006)

This effect is increased by strong temperature gradients. Areas close to the weld pool cool rapidly and shrink at a faster rate than those far away. The fact that the material is deposited in layers (with each step) makes the stress field even more complex, as the material that has been deposited is partly softened and re-shrunk by each new step. The interaction of these effects predetermines the appearance of a complicated stress pattern, leading to the warping, bending, twisting, or cracking of this material, particularly large components or low-ductivity thermally sensitive alloys like aluminium or low-ductivity steels (Ding et al., 2015b); (Frazier, 2014)

Heat input, speed of travel, bead height and cooling time are important factors that determine the severity of residual stresses. When the heat input is high the size of the molten pool is large, the cooling proceeds sluggishly and when solidifying, it undergoes pronounced thermal shrinkage. Poor cooling probability between layers will cause a build-up of thermal and expansion. These effects are usually mitigated by changes like controlled interlayer cooling, path planning and preheating the substrate (Wang et al., 2022)

Due to the inability of measuring residual stresses experimentally, numerical simulation is extensively employed in order to forecast stress development and determining problematic situations. The FEM models of thermo-mechanical processes are capable of mapping the stress distribution, assessing the distortion trends, and defining the process optimisation strategies (Montevecchi et al., 2016b) The simulations form useful instruments in assuring dimensional stability and structural reliability of parts produced by WAAM..

2.3 Finite Element Modelling WAAM

This is where one should mention Finite Element Analysis (FEA) which is one of the most valuable tools used to study the process of Wire Arc Additive Manufacturing (WAAM). Temperatures produced in the process of WAAM are very rapid and fluctuate with the arc motion, and as such, thermal and mechanical responses are not easily measured in real time. Due to this fact, simulation is a realistic and precise method of predicting the existence of

heat flow, temperature variations, and subsequent residual stresses without undergoing numerous physical experiments (Chiumenti et al., 2010); (Lindgren, 2006)

Under FEA, thermal cycles are detailed and the engineer can see the influence of input of heat, the speed of the travel, and cooling terms on the temperature field and the stress accumulation. The simulations are used to determine the process conditions to reduce distortion and defects. FEA is also less prone to trial-and-error testing, both of which save time and waste material resources, which is essential when working with massive metal components produced by WAAM (Montevecchi et al., 2016b)

During the contemporary studies, coupled thermo-mechanical models have become the norm in research since they enable both the prediction of temperature fields and stress development during the build. These models aid in optimisation of parameters and the fact that those components of WAAM must maintain structural performance needs, particularly in the sectors like aerospace, automotive, and marine where reliability is paramount (Frazier, 2014)

2.3.1 Boundary Conditions and Modelling of Heat Source.

In order to achieve realistic WAAM simulations, proper representation of the heat source and boundary conditions is of paramount importance. The electrical arc in WAAM is highly concentrated and should be represented mathematically in a way that reflects real behavior of the molten pool. Among others, the double-ellipsoidal Goldak heat source model is widely used because it divides the energy input into the front and rear regions, creating a realistic distribution of heat as the arc moves along the deposition path. This model has been independently utilized by (Gokuldoss et al., 2017) and (Honaryar et al., 2022) In addition, boundary conditions of the thermal type have a strong impact on the predicted temperature field. Cooling rate and stability of the molten pool are controlled by heat conduction into the substrate, convection to surrounding air, and radiation of heat at high temperatures. When these conditions are not defined correctly, there is a great chance of the simulation grossly overestimating or underestimating temperatures substantially. Other important factors include dwell time between layers, which controls heat accumulation between passes, and environmental conditions such as preheating or insulation of the base plate. When all those parameters are simulated correctly, FEA may closely reproduce real temperature gradients

and cooling behavior occurring during the WAAM process. This is according to (Graf et al., 2018a)

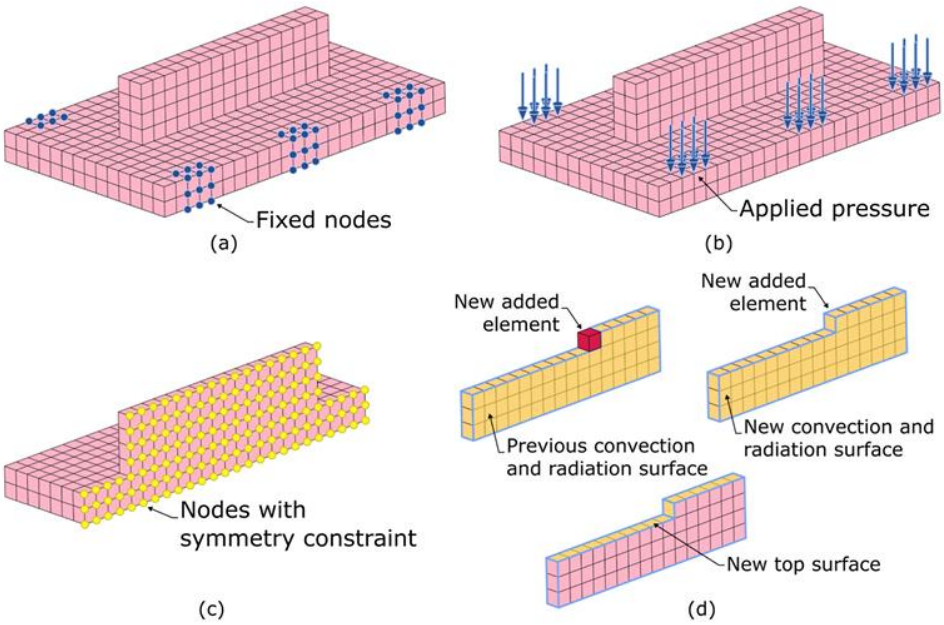


Figure 5 Boundary conditions in WAAM. (a) nodal, (b) surface and (c) symmetrical plane boundary conditions with (d) a schematic representation of the 'element (Sampaio et al., 2023)

2.3.2 Simulation of Temperature Fields and Thermal Cycles

Temperature field simulation is also necessary in the study of heat flow during the WAAM process and prediction of repeated heating and cooling of the material. As the electric arc is passed over the plate, the underlying metal is heated within a short time period creating a liquid pool with significant temperature differences between the centre of the weld and the solid material surrounding it. Goldak double-ellipsoidal heat source is a widely-used simulation model due to its realistic representation of this front-and-rear type of heat distribution, and its realistic representation of the shape of the typical WAAM molten pool (Galán Argumedo et al., 2024); (Honaryar et al., 2022) The cooling of the bead causes the heating of the next bead which in turn causes the heating up of the previous layer of the bead resulting in overlapping thermal cycles which define the total thermal history of the printed part. These cycles determine the rate of cooling, the scale of the area of impact of heating, and temperature distributions within the bead. Proper definition of boundary conditions like conduction to substrate, convection to surrounding air and surface radiation is significant since it has a direct impact on cooling behaviour. Without an adequate modeling of these

conditions, predictions of temperature can either be excessively high or excessively low and thus will cause errors in the mechanical simulation. Peak temperatures, thermal gradients, and cooling times can be realistically defined by FEM, and therefore, thermal simulation becomes a significant cornerstone to subsequent stress analysis (Graf et al., 2018a)

2.3.3 Stress Analysis and the Stratagem of validation.

The stress analysis is an important element of WAAM simulation since large temperature variations during printing, introduce a state of thermal expansion and contraction, which result into internal stresses after the material solidifies. Every deposited bead is expanded on heating and contracted on cooling but the contraction is limited by the solid substrate or already deposited layers. The resultant mismatch results in residual stresses that will result in distortion, bending or cracking once these stresses are too large (Lindgren, 2006) Stresses are computed using temperature fields of the thermal simulation coupled with temperature-dependent properties like elastic modulus, thermal expansion coefficient and yield strength to assess the thermo-mechanical FEM models. The activation of the elements is employed to give an emulation of the actual layer-by-layer deposition sequence and enhance the realism of the results thus assisting in prediction of stress concentration locations. Although numerical simulation does offer useful information, the findings have to be checked to make sure they are reliable. Through the use of experimental methods, which typically include thermocouples to monitor temperature, Digital Image Correlation (DIC) to measure deformation or X-ray diffraction or hole-drilling to measure residual stress, are typically compared (Montevecchi et al., 2016b) The model is a sure method of optimising the parameters of the WAAM process, minimising an undesired distortion, and enhancing the dimensional accuracy and structural stability of the printed components when the results of the simulations match the experimental measurements (Chiumenti et al., 2010)

3 Methods

This chapter explains the computation and simulation methods to be applied to study of the WAAM process. The methodology consists of the creation of a 3-D model, the use of temperature dependent material properties, boundary and thermal conditions definition, and the definition of a volumetric heat source to simulate the welding arc. A temporary thermal analysis was used together with a stationary structural analysis to recreate the thermal distribution and formation of stresses in the process of deposition of S235 structural steel.

3.1.1 3D Model and visualization

This simulation started by following the two steps which are common in the literature regarding the processes of welding and the use of WAAM. The steps are fairly similar to the ones adopted by previous works like those by (Montevecchi et al., 2016a), (Ding et al., 2015b) and (Zhang et al., 2023).

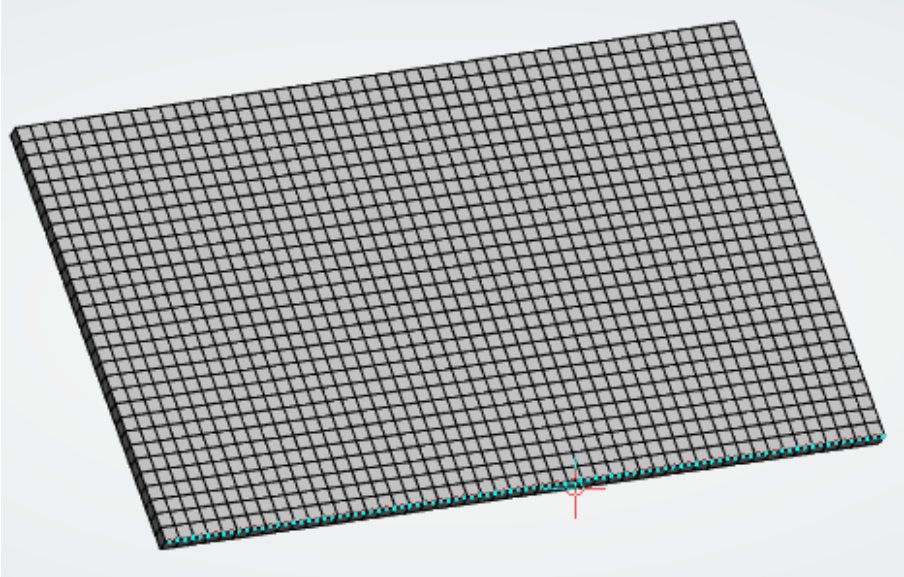
The first step included a transient thermal analysis carried out to simulate the motion of the source of heat, record the way temperature changes with time and space in the deposition process. The second step which was a static structural analysis involved the use of the resulting temperature field in which the field was applied as a thermal body load to compute the resulting thermal stresses, plastic deformation and distortion upon cooling (Dilberoglu et al., 2017).

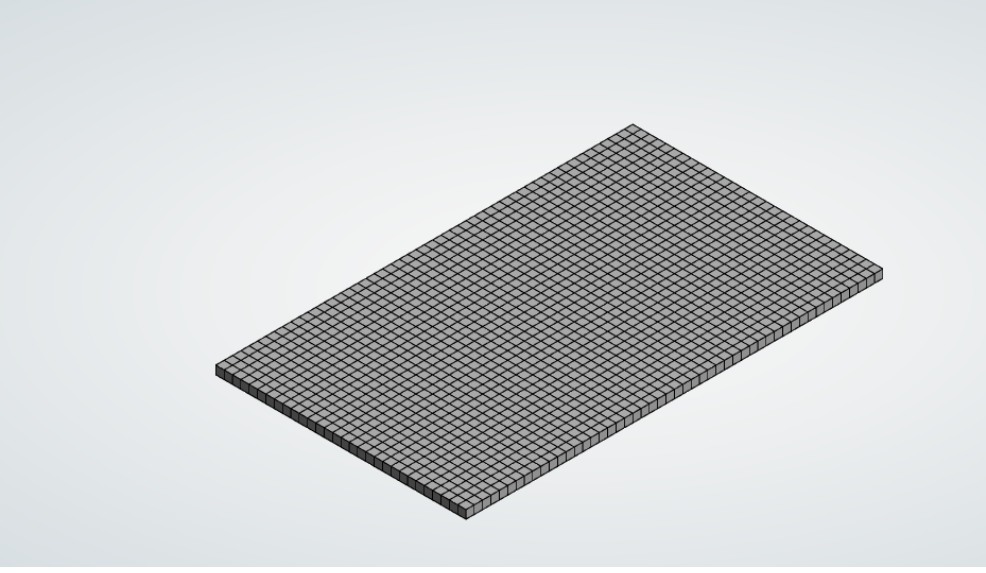
In order to make the model realistic, it contained material properties that were temperature dependent, realistic boundary conditions and Goldaks double-ellipsoidal heat source. This combined thermal-mechanical model allowed forecasting both the temporal variation in temperature and the mechanical university actions in the long term like residual stress and warping .

In general, this simulation system can be regarded as a credible online resource to investigate the WAAM process. It aids in the optimization of the process, assists in the validation of the results using experimental data, and supports the implementation of the dimensional accuracy and quality of additively manufactured components(Rodrigues et al., 2019).

3.1.2 Geometry and Meshing Strategy

Meshing strategy and geometry are necessary in obtaining stable and accurate finite element simulation in Wire Arc Additive Manufacturing (WAAM). This experiment utilized a single bead on plate design to recreate a single layer of material on a flat surface of S235 structural steel. This model gives an acceptable trade off between physically realisticbehavior and computational affordability. It enables the individual heat flow analysis, temperature gradient analysis, and the formation of residual stress (Montevecchi et al., 2016b), (Yan et al., 2025)





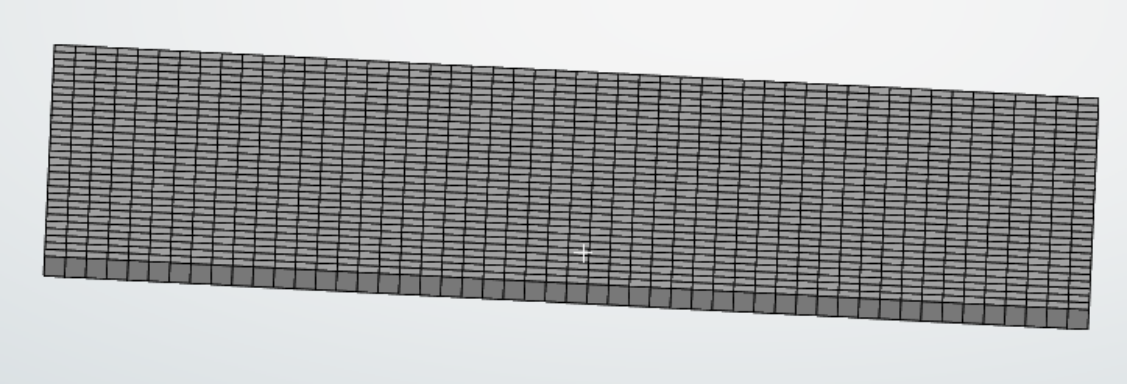


Figure 6 Finite element mesh of the plate

In SolidWorks the CAD geometry was designed and imported to ANSYS Workbench 2025 R2 where it was meshed and run. Certain geometric simplifications were used to decrease the computational effort required but still represent the most important thermal and mechanical behavior of the process. Figure 1 indicates that the hexahedral (brick) element in the form of a structured mesh was created. The mesh was optimized along the weld bead and heat-affected area (HAZ) so that accuracy will be enhanced in areas where there is a lot of thermal gradient and stress concentration (Codina, n.d.).

3.1.3 Model Geometry Design

The geometric model that will be employed in this study will be the one that has a rectangular base plate and a single deposited bead. The base plate has dimensions of 50 mm x 20 mm x 5 mm and the bead has a width of 5 mm and height of 3 mm. Such values are in agreement with typical WAAM simulation models of steels structure as given by (Montevecchi et al., 2016b) and (Zhang et al., 2023).

To make the model simpler, the bead was made a rectangular block whose surface was uniform rather than the curved surface that was normally created in real welds. Similar to (Ding et al., 2015b) and (Ahmad et al., 2020) this method will reduce the complexity of meshing and prevent a numerical instability, but still provide some good predictions of the heat flow and the temperature distribution.

Minor surface irregularities, rounded corners, and fillets were eliminated out of the model. (Yan et al., 2025) has stated that the elimination of these types of details has little effect on

the total heat transfer because the highest temperature gradients are found in the area of the weld pool and heat-affected zone (HAZ).

3.1.4 Coordinate System and Heat Source Path

The Cartesian coordinate system was introduced on a global basis to determine the motion of the source of heat and apply the right boundary conditions in the simulation. The directions of the coordinates were considered as follows:

The position of the heat source travel (longitudinal) is used as the X-axis.

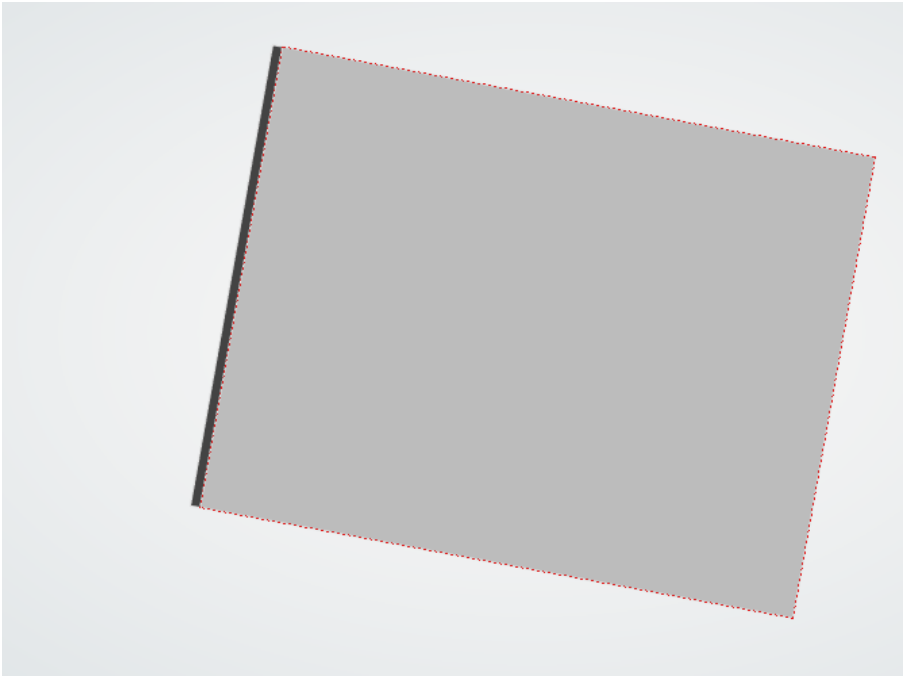
The X-axis is the perpendicular direction in relation to the width of the beads.

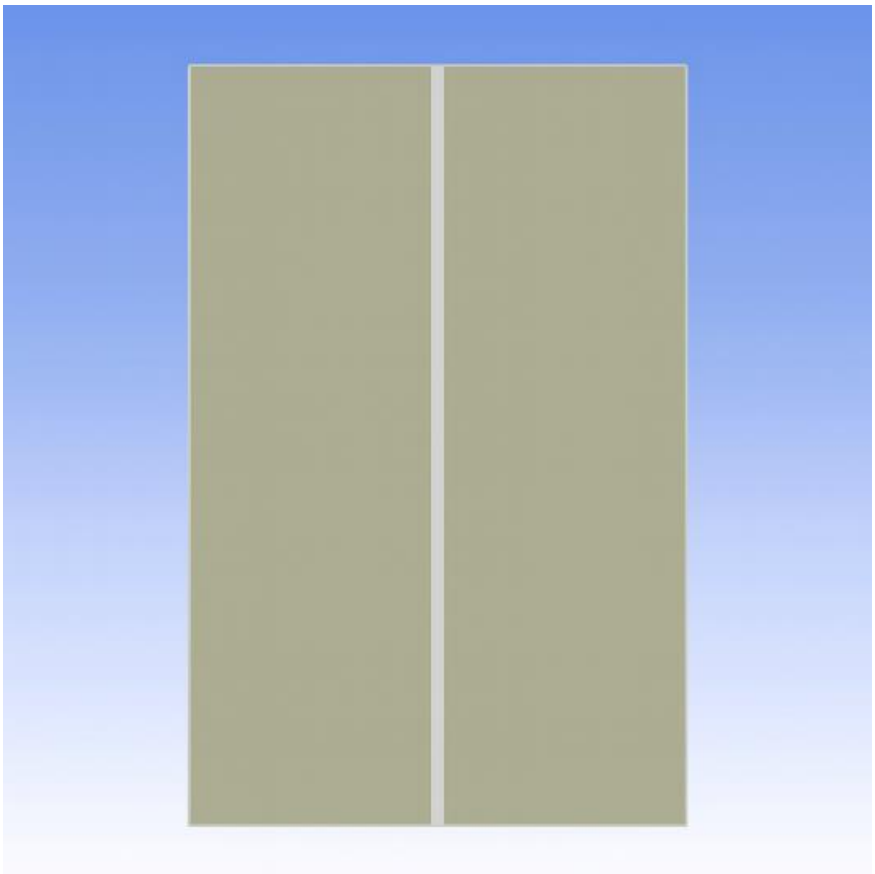
The height of the deposited layer is the Z-axis of the deposited layer.

The base plate corner that was on the bottom-left was the origin of the coordinate system. The movement of the heat source was characterized along the X-axis and the equation was:

$$x(t) = vt$$

where v is welding arc travel speed (m/s), and t is time (s). An example of a travel speed of 0.001 m/s was chosen and is in line with moderate deposition rates in the simulations of WAAM single-bead (Montevecchi et al., 2016b), (Belhadj et al., 2021)





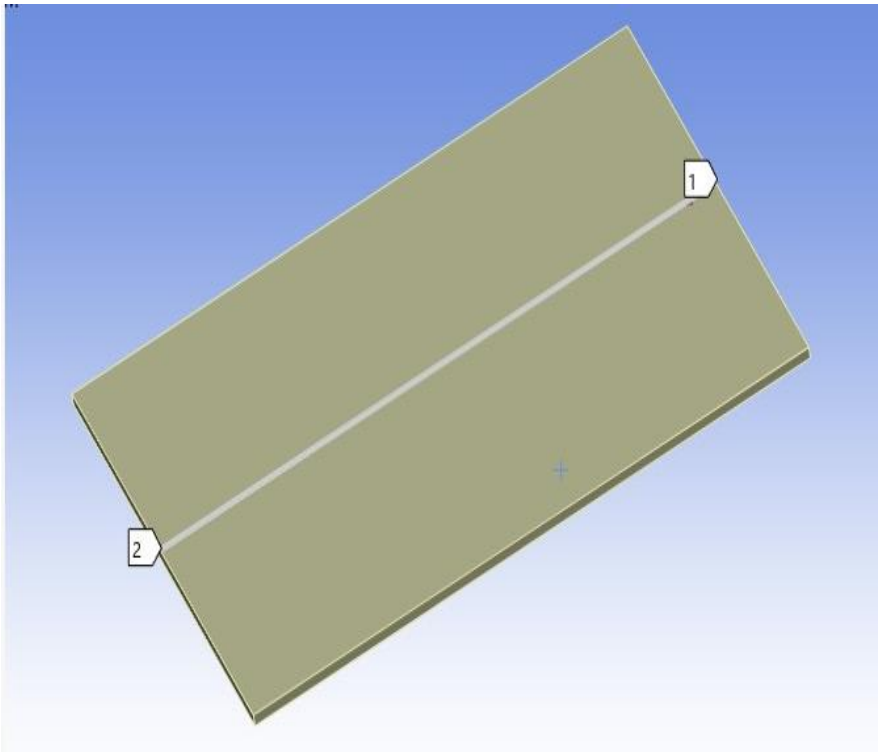


Figure 7 Geometry of the single steel plate with and without the weld path along the centrelin

The simulation in Figure 3.1 involved a single flat plate of S235 steel and the weld bead was deposited along the central axis of the plate. This arrangement provided a balanced and equalized heat input in the course of the simulation. The coordinate system defined was also important in the correct mapping of the Goldak double-ellipsoidal heat source and consistency in the thermal and mechanical analysis procedures.

3.1.5 Finite Element Meshing

To achieve a high fidelity of representation of the temperature and stress field in the Wire Arc Additive Manufacturing (WAAM) process, the model was discretized with hexahedral elements of 3D. In case of the transient thermal analysis, SOLID70 elements were employed. These elements have thermal simulation capabilities and can have temperature-dependent material properties. In the case of the stationary structural analysis, the elements used were SOLID185, which is capable of nonlinear material behavior and therefore can be used in stress and deformation analysis. This choice of terms is consistent with the applications of coupled thermal-mechanical modeling (Bauer et al., 2021).

An uneven grid was used to trade accuracy and computing power. As the temperature varies quickly in the weld bead and heat-affected zone (HAZ), a smaller mesh was employed with the most small element size of 1 mm. The mesh in the outer parts of the substrate was gradually enlarged to a final size of 4 mm. Such meshing approach gave a final number of 35,000 to 45,000 elements, which were determined by the mesh refinement parameters.

A mesh-independent study was done to get the accuracy of the solution. The successive convolution of mesh densities of the simulations was compared by the following convergence criterion:

$$E_{mesh} = \frac{|T_{i+1} - T_{i-1}|}{T_i} \times 100$$

WhereT_i are T_{i+1}. are the maximum temperature of two mesh sizes. When the difference between the mesh and the mesh was zero, this mesh was said to be converged when $E_{mesh} \leq$ 3%.

Also, the quality measurements of elements like aspect ratio and skewness were checked and should have been within the range of 0.85 as recommended by ANSYS of good-quality mesh. The quality of mesh proved useful in ensuring stable convergence of either the thermal or structural simulations (W. Zhang et al., 2023), (Ahmad et al., 2020).

3.1.6 Boundary Conditions and Thermal Constraints

The model of the finite element used realistic boundary conditions to model the thermal loss and mechanical constraints in the Wire Arc Additive Manufacturing (WAAM) process. These requirements were necessary to record the temperature development and structural distortions that occur with precision.

Thermal boundaries:

Both natural convection and thermal radiation were used to model the heat loss to the environment in the thermal simulation. An ambient air was subjected to the top and sides of the steel plate and the convection coefficient on the top and sides was set to 4-12 W/m 2 . -deg C depending upon the orientation of the surface. These values are representative of the common free-air cooling conditions and they agree with the past (Montevecchi et al., 2016b), (Andrade Tatemoto et al., 2024) The bottom surface, which was supposed to come into

contact with a support fixing, was associated with a lower value of the convection coefficient, indicating low levels of heat loss through conduction with a cooler heat sink (Murti et al., 1993).

$$q_{conv} = h(T - T_{amb})$$

$$q_{rad} = \varepsilon\sigma(T^4 - T_{amb}^4)$$

where $h = 8.5\,W/m^2.^0\,C, \varepsilon = 0.8, \sigma = 5.6 \times 10^{-8} W\,/\,m^2 \cdot k^4,$ and $T_{amb} = 22^0 C.$ Besides convective losses, the thermal radiation was also modeled with a surface emissivity of 0.8, which was assumed to be uniform across all the external surfaces. The external environment was taken to have a constant ambient temperature of 25 o C. The coupled convective and radiative boundary conditions made sure that the heat dissipation of the workpiece became realistic during and after the deposition.

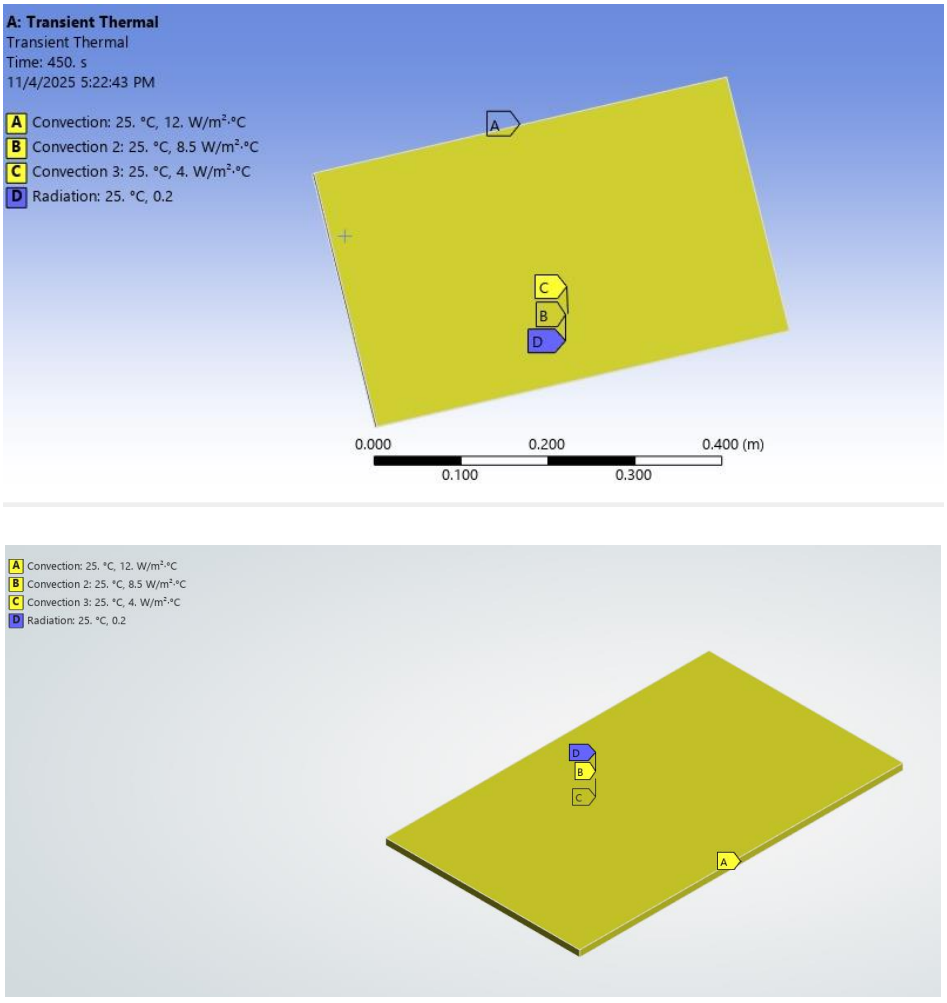


Figure 8 Thermal boundary conditions applied in the transient analysis, including convection and radiation

Mechanical boundaries:

In the case of the mechanical boundary conditions, a complete constriction of the bottom surface of the substrate in all the degrees of freedom was used to avoid rigid-body motion when analyzing the structure (Lambiase et al., 2025). The remaining surfaces were free to deform, and the processes of thermal expansion and contraction should take place naturally. This constraint model is the clamped state of a actual WAAM system, where the object under fabrication is firmly affixed to a base part all along the manufacturing procedure (Martina et al., 2012).

Initially, the simulation was started with all nodes at a temperature of 22 o C ambient temperature. During the X-direction travel of the Goldak double-ellipsoidal heat source along the weld path, transient heat transfer was calculated during the entire model. The transient thermal conduction governing equation was used:

$$\rho Cp \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q(x, y, z, t)$$

here $q(x, y, z, t)$ is proportional to volumetric power density of the moving Goldak heat source. Combining mesh refinement, realistic thermal losses and realistic clamping constraints allowed the solver to provide spatially and temporally resolved temperature fields which well represent the realistic thermal conditions in the WAAM process.

3.2 Material Models and Properties

In general, the accuracy of every thermo-mechanical simulation in welding or WAAM is determined to a great degree by the applied material model. Because of the large temperature range from room temperature up to above melting point in WAAM, most of the thermal and mechanical properties of metals change continuously with temperature, especially for steels. Therefore, material data dependent on temperature needs to be defined, such as thermal conductivity, specific heat, elastic modulus, and yield strength, in order to simulate realistic process behavior (Rodrigues et al., 2019).

In the present study, S235 structural steel was selected as the base material since it is a common low-carbon steel widely used in welded structures. The material was considered isotropic and homogeneous, as is commonly done at the macroscopic scale for structural steels. Temperature-dependent properties introduced to ANSYS Workbench via the

Engineering Data module were obtained and validated from published studies of (Montevecchi et al., 2016b), (Ding et al., 2015b), (Ahmad et al., 2020), and (Graf et al., 2018b) providing experimental and simulation-based datasets that describe the evolution of major thermal and mechanical parameters of steels during WAAM and related arc-welding processes. Incorporating these datasets ensures that the numerical model is able to represent realistic thermal gradients and mechanical responses under process conditions for S235.

3.2.1 Thermal Properties of Structural Steel

One of the areas of concern behind the behavior of structural steel in the welding process is thermal properties because it is an essential property that determines the rate of heat absorption, conduction, and storage. These properties strongly depend on temperature as a result of the alteration of the vibration of atoms as well as the activity of electrons. Thermal conductivity, specific heat capacity and density were the three important thermal parameters that are used within the simulation.

Thermal conductivity (k) is what controls the speed in which heat diffuses through the material. In the case of structural steel, it decreases with temperature, ranging between about 54 W/m k at 25 o C to around 25 W/m k at 1000 o C (Lindgren, 2006). This reduction assists in localizing the heat input in welding and the large part of the thermal energy goes close to the Cambridge of the weld pool which is vital in producing the correct thermal gradients (Lambiase et al., 2025).

Specific heat capacity is directly proportional to the temperature and is used in determining the energy needed to raise the temperature of one unit mass of material. It increases to approximately 470 J/kg -1K at room temperature to approximately 700 J/kg -1K by 1000 o C, meaning that the material holds more energy as it heats up (Montevecchi et al., 2016a). This conduct is particularly significant in the vicinity of the fusion zone (where vast sums of energy are taken on board in the melting stage).

The density (r) of the structural steel has also a slight temperature dependence that is not negligible. When the material heats up, its density decreases a bit--1000 degC: the material changes its density, approximately, to 7660 kg/m^3 (Andrade Tatemoto et al., 2024). Although the change is small, adding the density, which varies with temperature, will assist

in keeping proper predictions of the mass distribution and energy conservation during the thermal cycle.

An integrated combination of these thermal properties is used to guarantee the capability of the simulation to reasonably represent the dynamic behavior of heat transfer in WAAM. As thermal conductivity decreases and heat capacity increases with welding temperature, the heat diffusion rate is slowed, creating sharp temperature gradients important in predicting the extent of the melt pool, cooling rate and the subsequent creation of residual stress.

3.2.2 Mechanical Properties

Mechanical properties explain how the material reacts to thermal expansion and mechanical stress once the temperature field is known. For the structural steel, three temperature-dependent properties of importance were included in the model: Young's modulus, Poisson's ratio, and the coefficient of thermal expansion.

The Young's modulus, E, that essentially gives the stiffness of the steel, significantly decreases with an increase in temperature. The value of E decreases from around 210 GPa at room temperature to about 40 GPa at 1000 °C, as shown by (Lindgren, 2006). Such a drop reveals the softening of the steel upon heating, which has been an important basis on how material releases stress during the heating and cooling phases of WAAM.

The Poisson's ratio ν, representing the relation between axial and lateral strain, has been assumed constant due to the fact that its variation with temperature is small. A constant value of 0.3 was adopted, typical for low-carbon steels, and supported by experimental observations (Graf et al., 2018a; Rodrigues et al., 2019).

The following property, coefficient of thermal expansion (α), indicates the rise in the material with increase in temperature. For structural steel, α rises from approximately 12×10^{-6} /K at 25 °C up to approximately 18.5×10^{-6} /K at 1000 °C. This value is key to calculating distortion and residual stress build-up during cooling (Lindgren, 2006). These mechanical properties, when combined with thermal data.

These mechanical properties, when combined with thermal data, constitute a description of the coupled thermo-mechanical response of steel in welding or WAAM. Large temperature gradients generate strong local stresses due to thermal expansion, which counterbalance the

reduction of stiffness with temperature. It is this interaction which defines how stress and deformation will be developed throughout the process.

3.2.3 Plasticity and Yield Behavior

Plastic deformation occurs when thermal or mechanical stresses in the material are higher than its yield strength. In welding or WAAM, regions close to the molten pool often reach temperatures higher than the recrystallization point and this results in permanent plastic flow and residual strain. An elastic-perfectly plastic material model was implemented to capture material response beyond the elastic limit in this work. According to (Lindgren, 2006) the yield strength of structural steel decreases rapidly with temperature from approximately 250 MPa at 25 °C down to around 50 MPa at 800 °C. This behavior underlines that during heating, the strength of the material near the liquid pool is reduced, which is an important fact for understanding permanent deformation and redistribution of residual stresses during cooling (Kokare et al., 2024).

The total strain in the material can be expressed as:

$$\varepsilon_{Total} = \varepsilon_{elastic} + \varepsilon_{p_{lastic}} + \varepsilon_{thermal}$$

Where

$\varepsilon_{thermal}$ = stress-dependent elastic strain

$\varepsilon_{thermal}$ = irreversible plastic strain

$\varepsilon_{thermal} = \alpha(T - T_0)$ = strain caused by thermal expansion.

where ε_elastic is the stress-dependent elastic strain, ε_plastic describes the irreversible plastic strain, and ε_thermal = α(T - T_0) describes the strain due to thermal expansion. The above relationship enables the solver to calculate the total strain history at any node and therefore link the transient temperature field with the emerging stress resulting from the WAAM process.

3.3 Goldak's Double Ellipsoidal Heat Source Model

In simulations of welding and WAAM, the mode of heat input is very influential for the accuracy of the temperature results. Real arcs in welding do not distribute heat uniformly most of the energy is concentrated at the center of the arc and decreases slowly toward the outside. It is evident that such behavior cannot be properly described through simple surface heat flux, especially if the accurate depth of penetration, or shape of the molten pool as observed in the case of real welding, is to be correctly captured (Ning et al., 2020).

To make it more realistic, several models of volumetric heat sources have been proposed. The most used in welding and WAAM simulations is the Goldak Double Ellipsoidal Model mostly used in both welding and WAAM simulations (Gokuldoss et al., 2017). It provides a three-dimensional distribution of heat that can represent how energy is being deposited into the metal as the arc moves, hence giving better temperature and stress predictions (Montevecchi et al., 2016b) (Ding et al., 2015b).

The Goldak model is particularly suitable for arc-based additive manufacturing since it captures both the front preheating effect and the rear molten-pool behavior during motion of the welding arc. It provides this dual representation that allows for realistic thermal gradient simulation of the WAAM process and thereby enhances predictability of heat flow, material melting, and also stresses within the printed part.

3.3.1 Double Ellipsoid and Mathematical Representation of Heat Flux

In Goldak model of heat source the heat input in welding is divided into two crossing 3D ellipsoids, which move together along the weld line. The model shows in a realistic way how heat is dispersed during arc welding.

Front Half Ellipsoid:

Front ellipsoid is the zone immediately ahead of welding arc where the metal is being warmed up before melting. The heat input at this point is obtained as follows:

$$q_f(x,y,z) = \frac{6\sqrt{3}f_f Q_{eff}}{\pi\sqrt{\pi}abc_f} exp\left[-3\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{{c_f}^2}\right)\right]$$

Rear Half Ellipsoid:

The arc is called the molten pool and is represented by the rear ellipsoid, which contains the majority of the energy. As the liquid metal is located in the rear zone, a higher percentage of the total heat input is gathered in the rear zone. This results in an unsymmetrical temperature distribution such that the back part is longer and warmer than the front. This trend of heat distribution is supported by experimental research on the field of gas metal arc welding (Yan et al., 2025), (W. Zhang et al., 2023) The amount of heat input at the back part can be defined as:

$$q_r(x,y,z) = \frac{6\sqrt{3}f_r Q_{eff}}{\pi\sqrt{\pi}abr} exp\left[-3\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{{c_r}^2}\right)\right]$$

The local heat input is equal to the addition of an ellipsoid plus an ellipsoid:

$$Q(x,y,z) = q_f(x,y,z) + q_r(x,y,z)$$

The shape of each ellipsoid is (a, b, c), and the power ratio is (f_f and f_r) which enables the model to regulate both the depth of penetration and weld pool shape. The combination of these forms a constant and uninterrupted action of heating that flows along with the welding torch.

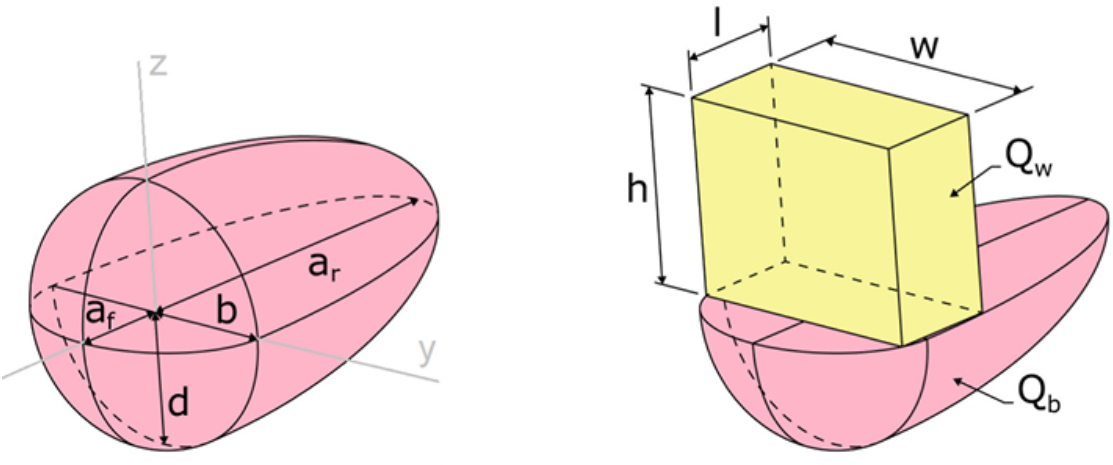


Figure 9 The double ellipsoid heat source model (Sampaio et al., 2023)

The rate of heat generation q(x, y, z) modelled by two Gaussian distributions one at the front and the other at the rear. These exponential functions can successfully model the banana-shaped heat zones in actual welds, and it has been established in previous numerical and experimental studies (Wu et al., 2009), (Montevecchi et al., 2016a).

3.3.2 Effective Power and Arc Efficiency

The overall amount of electrical power needed to power the process is determined as:

$$Q_{T_{otal}} = V \times I$$

The metal only absorbs a part of this power due to some of it being lost through the process of reflection and radiation. The fraction that is absorbed is given by the thermal efficiency (e).

$$Q_{eff} = \eta \times Q_{Total}$$

In the case of gas-metal-arc welding and gas-tungsten-arc welding, e is usually between 0.7 and 0.9 (Ding et al., 2015b). (Yan et al., 2025).

Definitions of the power fractions of the two ellipsoids are made such that:

$$f_f + f_r = 2$$

The most frequently used ratio is f f=0.6 and f r =1.4, which implies that the rear zone absorbs approximately 70 percent of the energy that is absorbed. The resultant ratio has a realistic temperature asymmetry in the simulated molten pool.

3.3.3 Effect of Geometric Parameters (a, b, c_f, c_r)

All the dimensions of ellipsoid affect the distribution of heat and the geometry of the weld-pool are:

Table 1 Ellipsoidal heat source parameters and their effects

Parameter	Physical Meaning	Effect on Temperature Field
a	Half-width along X-axis	Controls overall weld width (larger a refers wider, shallower pool)
b	Half-width along Y-axis	Affects penetration (smaller b refers deeper heat penetration)
$\boldsymbol{c_f}$		Governs pre-heat region ahead of arc
$\boldsymbol{c_r}$	Rear half-length	Controls trailing molten pool length (larger c_r refers longer tail)

f_r, f_f	Power fractions	Adjust symmetry of pool ($f_r > f_f$ gives banana-shape)

The most common parameter estimates of structural-steel WAAM were selected as a = 0.040m, b = 0.010m, c_r= 0.020m These parameters generate a long pool with realistic temperature gradients and smooth fusion boundaries, comparable to those of (Montevecchi et al., 2016b) and (Zhang et al., 2023).

3.4 Transient Thermal Analysis

Welding is associated with very localized heating, then rapid cooling and this gives rise to steep thermal gradients perpendicular to a weld and the underlying base material. Such gradients are the main source of the residual stresses deformations, and in the worst situations, cracking in the welded pieces (repository.lib.ncsu.edu). The proper prediction of these thermal effects would be critical to the structural integrity and dimensional stability of the assemblies welded together (Agelet De Saracibar et al., n.d.). To determine this, one will need to know the dynamic nature of temperature during the process.

Time-dependent heat conduction equation is the governing equation of transient heat transfer in the welding process, and it is written as:

$$\rho Cp \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q(x, y, z, t)$$

Analytical solutions are unpractical because of the complexity of this equation especially when the source of heat is moving, the geometry is inhomogeneous, and the properties are temperature dependent. Rather, the tube method of calculating the transient temperature field under realistic conditions is primarily used numerically by finite element analysis (FEA) (repository.lib.ncsu.edu). FEA provides the option of applying non-axisymmetric and 3-dimensional heat input models, and taking thermal boundary conditions and temperature-dependent material properties, and the results are quite more accurate than those found using simplified analytical methods (Yang et al., 2017).

One of the common models that can be applied in welding simulations is the double-ellipsoidal heat source as provided by Goldak et al. This is a thermal model that refers to the unsymmetrical temperature distribution about a traveling arc and can be used to simulate heat processes used during arc welding with great accuracy (Wu et al., 2009). As used in

FEA programs, including ANSYS or Abaqus the Goldak model dramatically increases the accuracy of thermal field predictions in welded materials (Murti et al., 1993).

3.4.1 Goldak's Double-Ellipsoid Heat Source Model and Time Implementation

In this work, to simulate the heat input of the welding, the model of the double-ellipsoidal heat source was used, as was initially introduced by (Wu et al., 2009). Currently, it is one of the most standard models used in FEA for modeling arc welding processes because of its great capability in closely simulating the space of the heat distribution during the welding. The model further subdivides the heat input region into two overlapping three-dimensional ellipsoids, one in front of the arc and the other in the back. With this asymmetric setup, the simulation is better able to capture the difference in thermal distribution between the front and back edges of the molten pool (DebRoy et al., 2018). The total heat input, Q is defined with relation

$$Q = \eta VI$$

η is a thermal efficiency of a process. In this simulation, the efficiency factor was taken to be 0.85, which is a normal efficiency factor in gas metal arc welding. According to the standard convention, it was 60 percent of the total heat input to the front ellipsoid and 40 percent to the rear.

The heat flux in every ellipsoid was represented with the three dimensional Gaussian:

$$q(x, y, z) \propto \exp\left(-3\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right),$$

Here, a, b, and c symbolize the longitudinal direction (travel direction), transverse direction (width) and through-thickness direction (depth) semi-axes, respectively. The parameters values were selected so that they represent the geometry of the desired weld pool.

Specifically, the longitudinal extents assigned to the front and the rear ellipsoids were different to reproduce the higher concentration of heat input in front-of-the-arc and the lower concentration of heating in the rear. Assuming that experimental determination of the weld bead geometry has been made, the parameters would have been calibrated closer to observed dimensions of fusion zone.

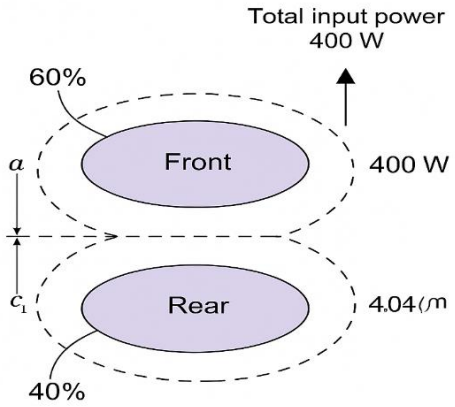
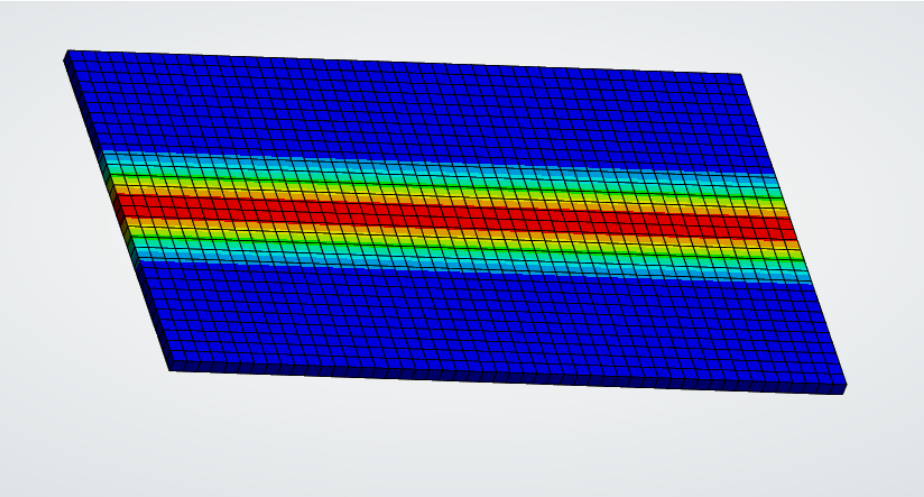


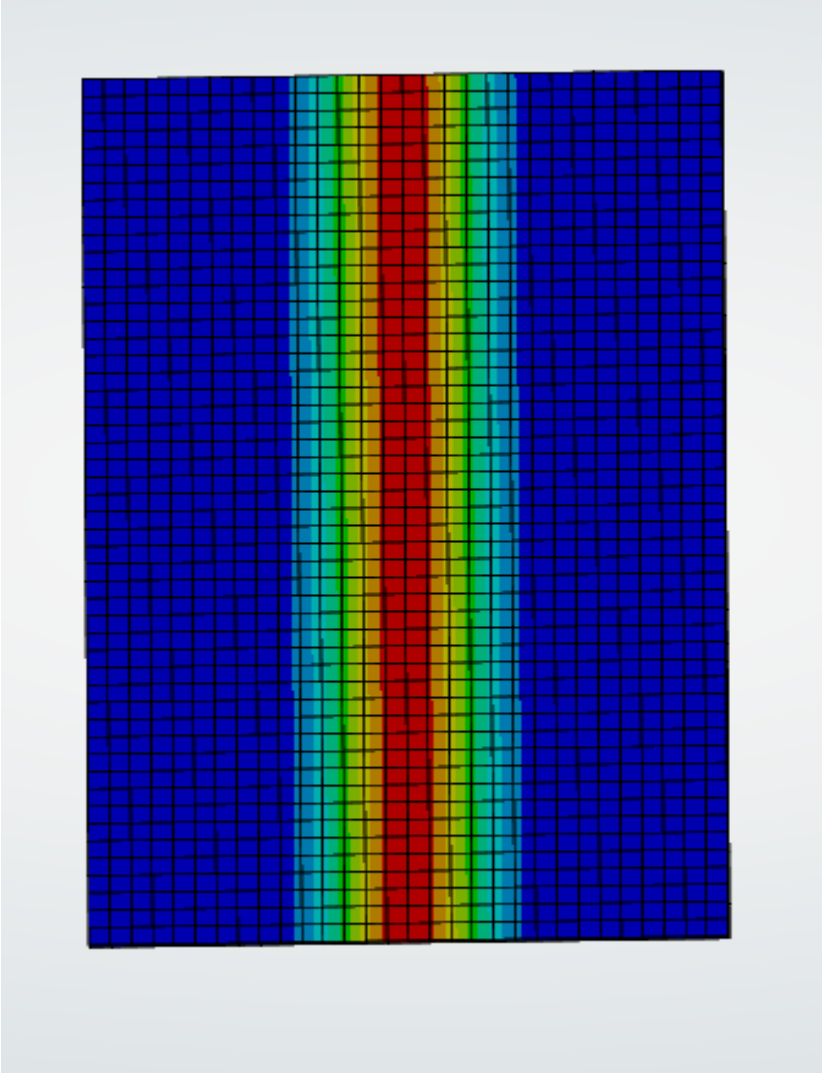
Figure 10Figure 5 Schematic of Goldak's double ellipsoidal heat source model showing front and rear heat distribution

Figure 3.4 shows the conceptual image of the model of the heat source that is double-ellipsoidal used in this research as it was initially proposed by Goldak et al. According to this model the volumetric heat input is separated into two spatially overlapping ellipsoidal zones, one in front of the center of the arc, the front ellipsoid, and one behind the arc, the rear ellipsoid. The asymmetric design allows to more accurately simulate arc welding, with a more focused thermal concentration at the arc front and the wider heat distribution at the back, where molten metal starts to solidify (Wu et al., 2009; Montevecchi et al., 2016b).

A heat input of 400 W was made effective in the present simulation and the thermal efficiency was 1 with a heat input. η) of 0.85. This ratio indicates the ratio of electrical energy, which is transformed into useful thermal energy, absorbed by the workpiece, a standard range of efficiency of gas metal arc welding processes (Ding et al., 2015a). Based on the common formulation of Goldak, the power input was divided to give the front ellipsoid 60 percent of the total heat and the rear 40 percent. This split is similar to physical welding in which a more intense and localized input of energy is provided by the arc front with the heat on the rear zone having been solidified and cooled.The ellipsoidal regions are mathematically determined by 3D Gaussian distribution functional which is defined by certain semi-axis. , b, c_1, and c_2The arc length, width and depth of the front and rear zones respectively. Parameters that were selected in this study were: a=0.055 m,b=0.0010 m,c1 =0.050 m,c2=0.050 m These values were chosen to fit an expected weld bead geometry and obtain a realistic depth to width ratio in line with operations of single-pass WAAM when using structural steels (Yan et al., 2025; Zhang et al., 2023). The specified ellipsoidal heat

source therefore provides a time and place dependent heat input that is capable of recreating the temporal and spatial thermal field that is found during arc deposition.





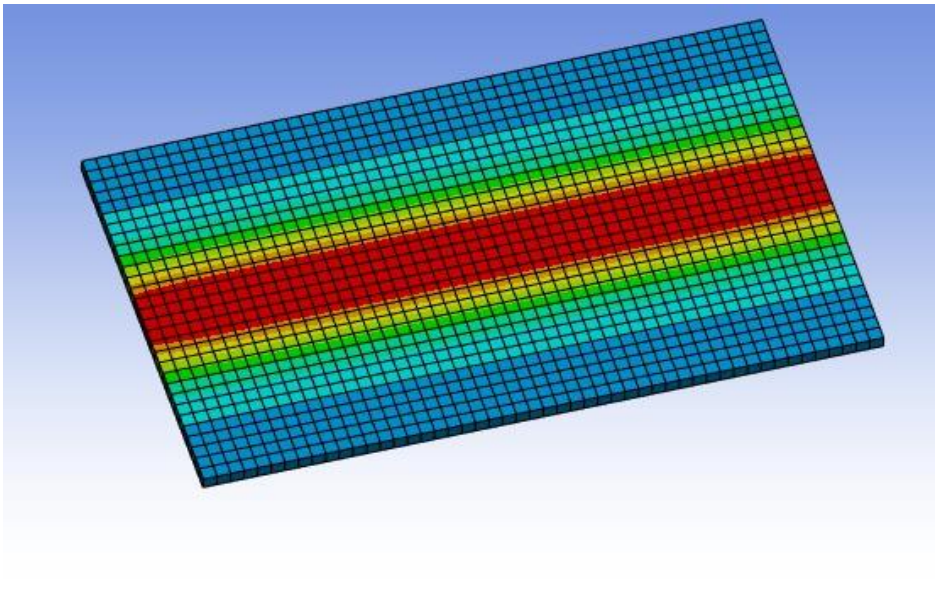


Figure 11:Temperature distribution during welding process

The heat source was set to move along the Z-axis, with a welding speed of 0.006 m/s, which characterized a uniform deposition process throughout the substrate. This motion was coupled with the Goldak double-ellipsoidal heat source model so that the prescribed volumetric heat flux interacted continuously with the mesh elements in the HAZ as the arc moved forward. Because the front and rear ellipsoids partially overlap in space, the thermal profile that resulted was of a smooth, continuous heat distribution which evolved both temporally and spatially in the simulation.

The simulation results showed that such ellipsoid overlapping produced an asymmetric, banana-shaped molten pool. This is typically expected from the thermal footprint created in arc welding processes and agrees well with the empirical observations made in both GMAW and WAAM on weld pool geometry. This shape naturally arises because of the nonsymmetric energy distribution in the Goldak model: the front half provides sharp preheating, while the rear maintains a broader thermal influence due to molten metal retention and solidification lag.

The capability of the simulation to reproduce this transient, asymmetric heating pattern confirms the correctness and reliability of the heat source implementation. Further, it assures that this model is able to simulate such complicated thermal cycles as in the real welding processes and is, therefore, quite applicable in the prediction of thermal gradients, stress formation, and microstructural evolution in WAAM applications according to Montevecchi et al. (2016b) and Wu et al. (2009).

3.5 Static Structural Analysis

In this work, after the transient thermal simulation, a static structural analysis was performed, with the aim of analyzing the mechanical response of the steel plate caused by the non-uniform heating and cooling generated in the process of welding. The temperature distribution obtained through thermal analysis has been imported directly into the structural solver as a thermal body load. This step enabled the proper simulation of material expansion and contraction with respect to time due to temperature changes (Ahmad et al., 2020).

This structural analysis has been conducted to determine the residual stresses that result, including plastic deformations and strain fields in the welded plate (Huang et al., 2025). All these mechanical effects act through thermal gradients: a heated material expands but its expansion is constrained by cooler parts around it, so that internally stresses are created. While cooling down, contraction takes place which can leave the weld zone with tensile residual stresses and areas around it with compressive stresses(Kotadia et al., 2021).

These various patterns of the state of stress and deformation need to be understood, as they markedly influence dimensional accuracy, structural integrity, and performance of the welded components. The detailed and accurate prediction of residual stress and distortion therefore assumes great importance for the improvement in weld quality and prevention of defects such as warping, cracking, or fatigue failure from occurring in an industrial context.

3.5.1 Thermal Loading and Boundary Constraints

The temperature distribution from the transient thermal simulation was mapped directly onto the structural model as a thermal body load. By using this direct method, the full time-dependent thermal history due to the welding process was captured, so that accurate simulations of stress and deformation due to a non-uniform temperature change were obtained (Herzog et al., 2016). Each node on the model received its corresponding temperature value over the entire simulation period of 450 seconds, including both the heating and cooling phases of the weld cycle.

Structural boundary conditions were applied in such a way as to prevent rigid-body motion, while allowing for the natural thermal expansion and deformation to take place. Namely:

- One edge of the plate was fixed in the X-direction to simulate typical clamping used in welding experiments, (Lindgren, 2006)
- A single node was constrained in the Y-direction to eliminate free movement without introducing artificial stiffness.
- The remaining surfaces were free, hence the plate was allowed to bend and also expand in a thermal situation.

This boundary condition set approximately models the realistic welding scenarios where local heating will cause expansion and upward distortion near the weld bead, while the cooler surrounding material develops compressive stresses. The interaction of thermal input with mechanical restraints allowed the model to realistically predict residual stress development and final deformation patterns observed in welded S235 steel plates.

3.5.2 Thermal Expansion, Stress Formation, and Plastic Deformation

Heat input from the welding source, moving along the plate, causes localized heating of the molten pool and HAZ, therefore causing thermal expansion in these regions. However, its expansion is constrained by the surrounding cooler material that cannot deform as freely. Consequently, internal stresses arise within the workpiece. Upon cooling, the regions which had previously been heated will shrink, and residual tensile stresses will develop at the weld center while compressive stresses are developed in areas outside the weld zone (Yang et al., 2017). These residual stresses are significant because they affect the structural performance of the material after welding.

The total thermal strain developed in the material due to temperature changes is given by:

$$\varepsilon_t = \alpha(T - T_0)$$

α = the coefficient of thermal expansion T = the current temperature, ε_t is the thermal strain and T_0 is the ambient reference temperature, and the equation describes just how the material expands upon heating and contracts upon cooling, causing stress and distortion.

To capture this mechanical behavior accurately, a temperature dependent plasticity model was adopted. More precisely, a bilinear isotropic hardening model was considered, allowing the material to yield and harden after reaching its ultimate elastic limit. In fact, one of the

important parameters in predicting plastic deformation is the temperature dependency of the yield strength in structural steel like S235(Belhadj et al., 2021). It was assumed that the variation of yield strength with temperature was given by 250 MPa at 25 °C, 220 MPa at 200 °C, 150 MPa at 400 °C, 80 MPa at 600 °C, and 30 MPa at 800 °C. This substantial reduction of yield strength at higher temperatures allows the model to capture material softening near the weld, particularly in the molten and HAZ regions, where plastic flow is expected (Kotadia et al., 2021; Murti et al., 1993).

Further, Young's modulus was considered as 2.0× 10^11 Pa, while Poisson's ratio was taken as 0.3. These properties enable the model to clearly capture the elastic behavior in cooler zones and plastic flow in high-temperature regions close to the weld pool during deformation processes. Simulation of both types of deformations is critical to predictions related to the residual stress field and final distortion in the welded plate. This will be useful for the assessment of mechanical long term behavior and reliability after the end of the welding process.

4 Results and Discussion

The numerical simulation results detail the thermal and mechanical behaviour of the WAAM process in the case of single-bead deposition on S235 structural steel. The transient thermal and structural analyses provide insight into the evolution of temperature during the passage of the arc and the generation of residual stresses during cooling. Such findings are important in understanding how the heat flow and thermal gradients lead to the development of stresses-a very important consideration in enhancing the quality and structural stability of WAAM-manufactured components.

4.1 Temperature Distribution During Deposition

The temperature field obtained with thermal simulation was highly non-uniform, representative of the arc-based additive manufacturing method. The temperature in the vicinity of the centre of the bead reached about 1586 °C at the peak of the heating cycle, which corresponds to the melting range for low-carbon structural steel. The temperature drops drastically with distance from the molten pool, going down to 195-543 °C in the zones surrounding it, and rises to near-ambient level at the far edges of the plate. This steep gradient reflects the concentrated heat input of the electric arc and is in good agreement with the thermal behaviour predicted by the Goldak double-ellipsoidal heat source model widely used in WAAM numerical simulations (Montevecchi et al., 2016b; Safyari et al., 2024).

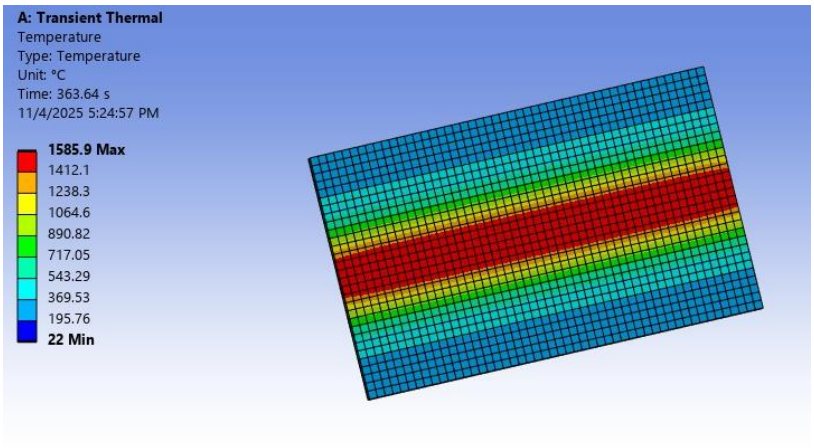
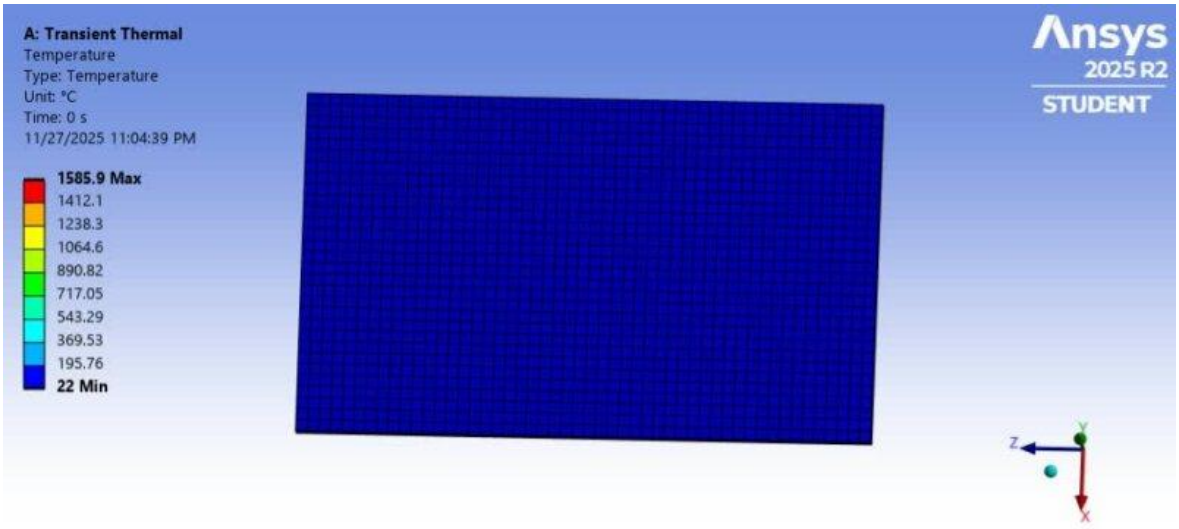


Figure 12: Temperature contour after the full welding cycle

The temperature distribution was relatively narrow during the early deposition stage, where only a warm band was developed along the path that the bead traveled, while the rest of the plate was little affected. During the deposition process, this warm area expanded gradually due to the conduction into the substrate and the continuous action of the moving heat source. At the middle of the heating cycle, the thermal field had a much wider span of peak temperatures within the range of 700-1200 °C. This tendency gives an indication of the HAZ that also starts to grow at this stage in most experimental and numerical WAAM studies.

Once the pool had reached its maximum extent, the peak temperatures had well exceeded 1400 °C over much of the central bead area. The molten zone took the form of an elongated high-temperature region, which mirrored the usual banana-shaped geometry characteristic of arc welding and WAAM processes. This thermal footprint is in good agreement with the behaviour documented in (Safyari et al., 2024), where the thermal field has elongated along the travel direction due to the accumulation of heat in the material. Similarly, for the WAAM videos presented in that work, the molten pool developed an elongated shape, further confirming that the expected heating pattern is well captured by the present simulation.



Video 1 Temperature distribution during WAAM from finite element simulation.

Toward the later stages of the heating cycle, the thermal field exhibited increased uniformity along the bead centreline, with a broad region of the plate reaching intermediate-to-high temperatures in the range of approximately 700 °C to 1500 °C. This widening band of elevated temperature reflects the progressive accumulation of heat as deposition proceeds. The simulation video included in this study further demonstrates this behaviour dynamically, showing how peak temperatures gradually shift and intensify along the travel direction,

confirming the presence of cumulative thermal build-up ahead of the heat source. This is typical of in-process heat treatment, whereby repeated exposure to the moving arc causes minor thermal build-up even under single-bead deposition (Safyari et al., 2024). similarly identified the same effect in multi-layer WAAM, where each new layer experiences a slightly elevated baseline temperature due to heat retained within the previously deposited material.

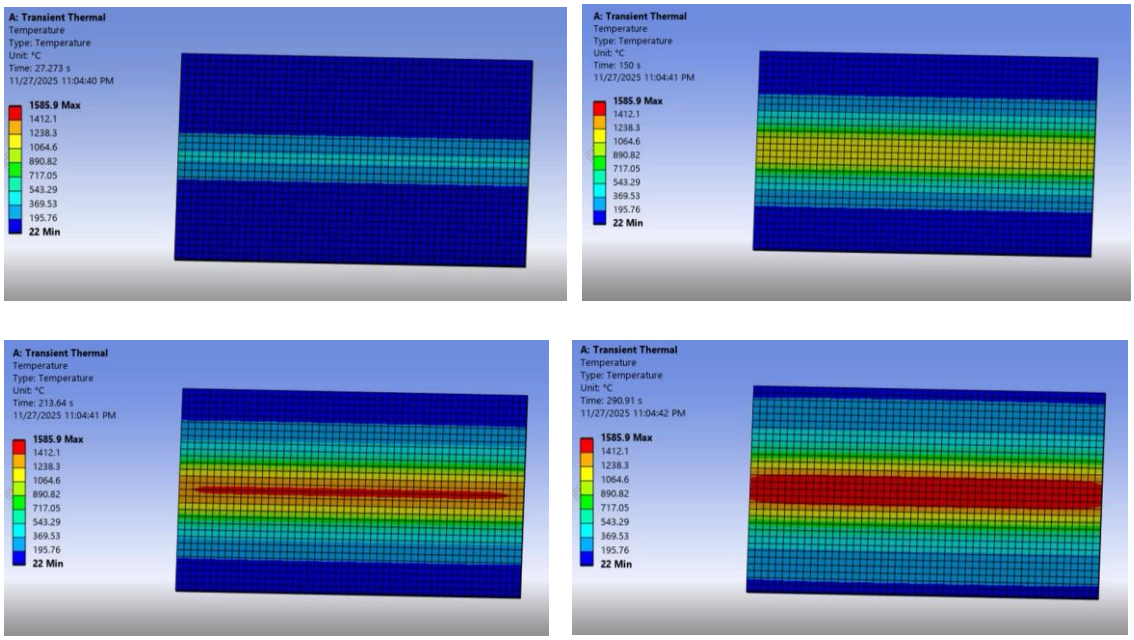
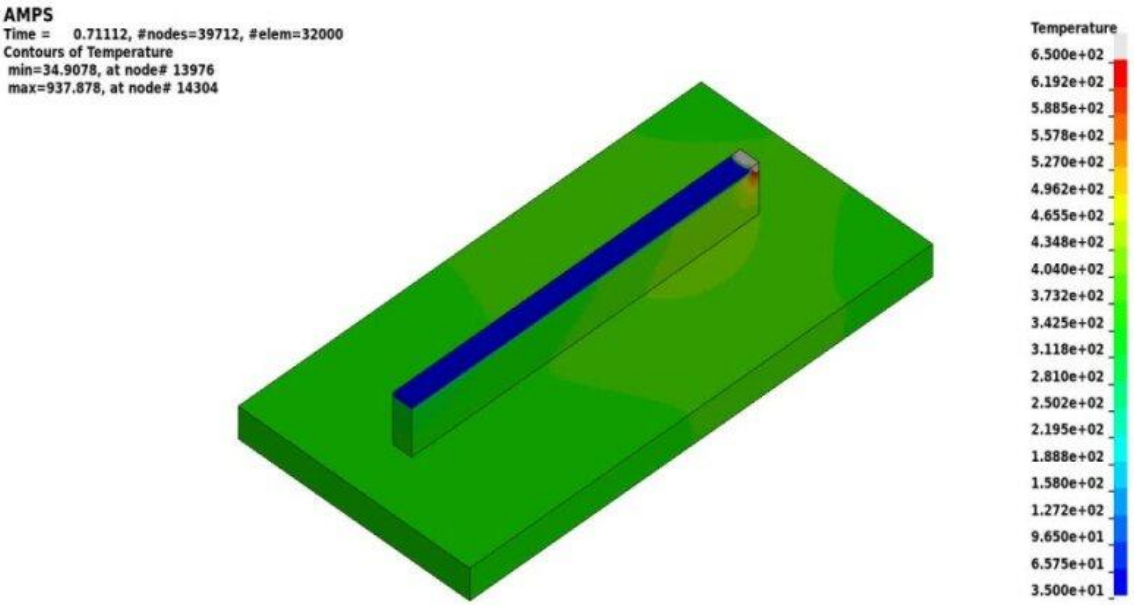


Figure 13 Time-dependent evolution of temperature distribution during single-bead WAAM deposition.

The thermal cycle was found to be of a transient nature during the analysis of the heating and cooling of every point on the plate. Each of them went through extremely fast heating when the arc approached and cooled down sharply after the heat source had moved further ahead. Although only one bead was deposited, the repeated heating and fast cooling pattern presented a close resemblance to the thermal cycles in multi-layer WAAM processes as shown in figure below. Research like (Andrade Tatemoto et al., 2024; Montevecchi et al., 2016b; Safyari et al., 2024) report similar short thermal cycles that are critical in the development of residual stresses and distortions addressed in the next section.



Video 2 Finite element simulation of temperature distribution during WAAM of Multilayer (all temperatures are in °C) (Safyari et al., 2024)

In summary, the behaviour that was recorded throughout deposition confirms that the simulated heat-flow model effectively captures the essential physical features of WAAM thermal behavior. The temperature evolution, the shape of the molten pool, and the width of the HAZ all correlate well with established research, where progressive heat accumulation provides a reliable basis for analyzing the resulting mechanical response.

4.2 Stress Evolution and Residual Stress Patterns

The structural simulation showed that the cooling of the deposited bead under contraction developed large magnitudes of residual stresses. Figure 7 presents the von Mises stresses, which had a maximum value of 3.5433×10^8 Pa ≈ 354 MPa and a minimum value of about 2.046×10^7 Pa. These values fall within the range for most WAAM steel parts, with maximum tensile residual stresses sometimes reaching 300-500 MPa for different heat inputs and material thickness.(Martina et al., 2012); (Rodrigues et al., 2019); (Wang et al., 2022). Such an agreement demonstrates that the coupled thermo-mechanical model successfully

captures the key deformation mechanisms related to bead-on-plate WAAM.

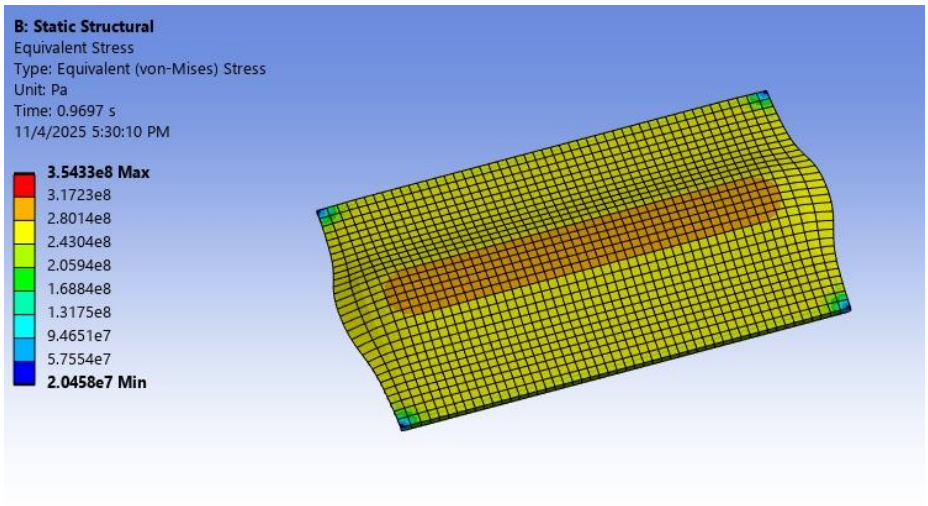
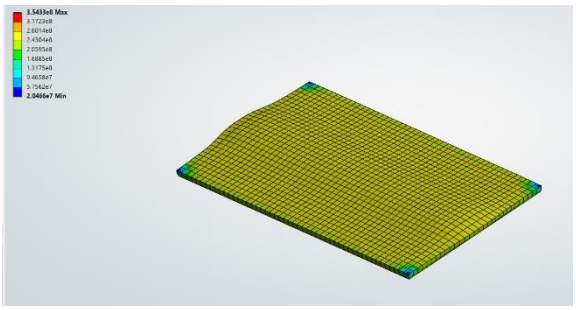
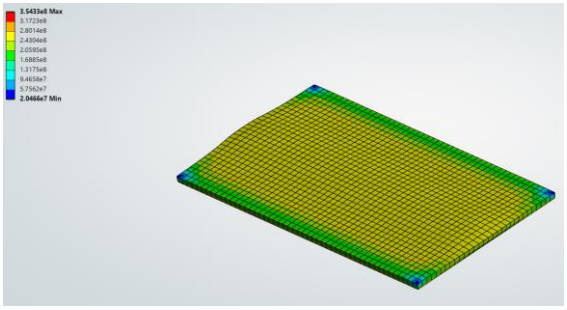


Figure 14: Equivalent (von-Mises) stress distribution

A closer examination of the time-dependent stress evolution further illustrates this behaviour. At t = 0.2 s, the stress field is still developing, with moderate tensile stresses appearing near the partially solidified region and low compressive stresses near the clamped edges. Peak stresses at this moment reach approximately 2.67×10^8 Pa, concentrated around the hottest portions of the bead. As cooling progresses to t = 0.4 s, the stress distribution becomes more defined, and a wider, more continuous tensile band forms along the bead path, increasing the stress level to nearly 2.81×10^8 Pa. By t = 0.7 s, most of the bead has solidified, and thermal contraction significantly intensifies the stress field, with values approaching 3.0×10^8 Pa as shrinkage becomes increasingly constrained by the substrate. When the material reaches t = 1.0 s, it has cooled close to room temperature, and the final residual stress state is established. At this point, the maximum stress reaches 3.543×10^8 Pa, forming a continuous high-tension region along the top surface of the bead





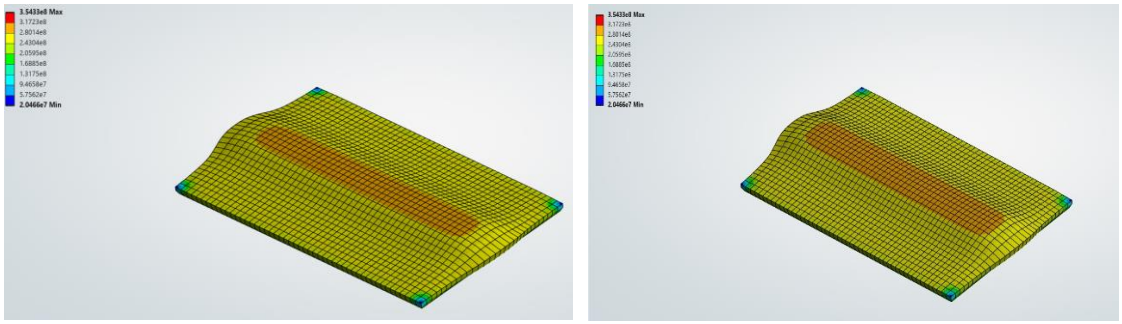
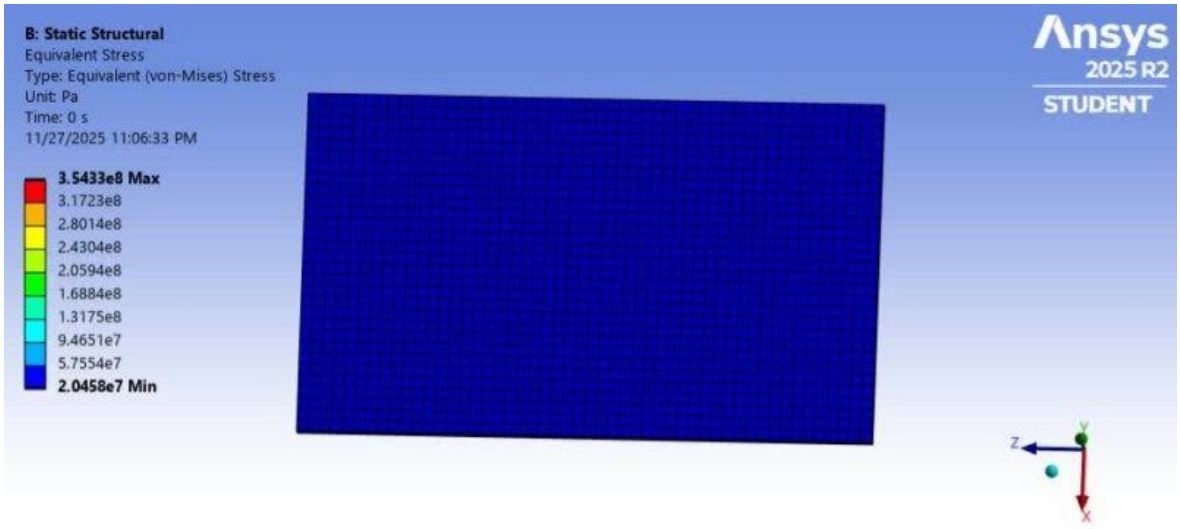


Figure 15 Time-dependent evolution of von Mises stress during the cooling of the deposited bead

The progressive build-up of this stress follows classical thermo-mechanical shrinkage mechanisms, described in welding simulations where the material is essentially stress-free at high temperatures but rapidly accumulates tensile stresses during solidification and cooling (Graf et al., 2018a; Rodrigues et al., 2019). The largest residual stresses develop at the top surface of the bead. Three physical mechanisms explain this observation. First, the top surface has seen the largest drop in temperature and consequently has large thermal strain gradients. Second, this is a single-bead deposition, meaning there is no reheating of the top layer from subsequent layers. Third, in multi-layer WAAM, subsequent layers tend to partially relieve stresses in earlier layers by thermal softening. This beneficial effect is absent here. Thirdly, the top region is less mechanically supported and thus sees the greatest tensile strain concentration due to contraction.



Video 3 Time-resolved finite element simulation of stress distribution during single-bead WAAM

These observations align well with the multi-layer simulation video presented in the (Safyari et al., 2024), where previous layers undergo stress relief due to reheating cycles, while the surface always has the maximum tensile stress. The video in (Safyari et al., 2024) shows the

thermal cycling and mechanical relaxation of underlying layers during multi-pass deposition and confirms that the final layer always exhibits the largest locked-in stress. The present single-bead simulation reproduces this same behavior but without the stress-relieving effect of subsequent layers.

The residual stresses across the plate thickness were relatively uniform and showed only a mild increase near the upper region of the bead. While no through-thickness stress plot is here, the pattern that was observed is consistent with findings from (Montevecchi et al., 2016b), who found limited through-depth variation in WAAM steel plates due to the shallow arc penetration and gradual cooling in the substrate. In this study, the heat-affected region extended for only a few millimeters below the bead while deeper regions cooled relatively uniformly, providing a stable and mostly constant stress field across the thickness.

Cooling time was also found to have a strong influence on residual stress formation. Longer cooling periods allowed for a more gradual contraction of the bead, reducing steep thermal gradients and lowering final stress levels. In contrast, rapid cooling locked in much higher stresses due to sudden thermal shrinkage. Such trends are documented, in parameter-sensitivity studies by (Montevecchi et al., 2016b; Wang et al., 2022), which all concluded that cooling-rate control is one of the major factors for WAAM process improvement relating to reduction of distortion and residual stresses.

Finally, boundary conditions provided important contributions to the final distribution of stress. The bottom face of the plate was fixed in order to simulate rigid clamping, a common practice in most real WAAM setups. This constraint prevented large global deformations, forcing the thermal shrinkage to be transformed into internal stresses instead of plate bending. Upon releasing the clamp, a small "spring-back" effect was seen, consistent with the literature related to welding mechanics (Williams et al., 2016b; W. Zhang et al., 2023). Although the fixed constraints increased the magnitude of residual stresses, they provided a realistic representation of the industrial WAAM conditions.

The good agreement of the present simulation with the multi-layer video in the reference paper and with published numerical and experimental studies attests that the adopted FEM model is able to predict the thermo-mechanical behaviour of WAAM steel accurately. Results demonstrate the proper spatial distribution, time-dependent evolution, and

magnitude of residual stresses, furthering the validity of the model for single-bead WAAM analysis. (Ning et al., 2020).

4.3 Effect of Process Parameters

The process parameters that have a great effect on the behaviour of the WAAM process include the most important heat input, travel speed, and the interlayer dwell time. These variables determine the amount of heat that enters the material and the way that heat is transferred into the material by conduction, convection and radiation. Consequently, they have a direct relationship with the size of the molten pool, the width of the heat-affected zone (HAZ), the thermal gradients during cooling and finally the magnitude and distribution of residual stresses. The current simulation trends are comparable with the past researches like (Ding et al., 2015b; Montevecchi et al., 2016b; Safyari et al., 2024; S. Zhang et al., 2021), All of these point out the sensitive relation between thermal parameters and mechanical results in WAAM.

Heat Input

Among all variables, the most influential is heat input, defined by arc power and wire feed rate. Increased heat input yields a larger molten pool, with higher peak temperatures and deeper substrate penetration. Steeper thermal gradients during cooling will result in higher residual stresses. This effect is well replicated in the simulation, as high-power deposition gave peak temperatures near 1586 °C and residual stresses in excess of 3.54×10^8 Pa. Higher heat input also serves to increase the effective width of the HAZ, leading to more severe thermal expansion and contraction cycles.

These findings are in line with (Montevecchi et al., 2016b), who showed that increasing heat input in WAAM builds serves to significantly heighten distortion and stress.. (Lambiase et al., 2025; Safyari et al., 2024) Further, they reported that excessive heat accumulation caused microstructure instability in multi-layer steel WAAM as a result of repeated reheating. On the other hand, reduced heat input can lower thermal gradients and stabilize the mechanical response; however, caution is warranted to avoid incomplete fusion or weak bonding.

Travel Speed

Travel speed affects the duration of exposure to the heat source and, thus, the effective heat input per unit length. If total heat input per length is maintained constant, travel speed by itself has little effect on final stress level - an observation consistent with (Ding et al., 2015b). But when arc power is not tuned and the travel speed changes, the effect becomes much stronger.

The consequence of higher travel speeds is less energy being deposited per unit length, with correspondingly smaller molten pools, reduced peak temperatures, and narrower HAZ; residual stresses tend to be lower under such conditions. Lower travel speeds mean that more heat is deposited per unit length, larger molten pools, and higher peak temperatures result in higher tensile residual stresses. This trend supports the conclusion of (Montevecchi et al., 2016b) it is the ratio of arc power to travel speed that is the dominant factor governing heat flow in WAAM.

Interlayer Dwell Time / Cooling Time

Although this study analyses only one bead, the influence of the dwell time is crucial to understand general WAAM behaviour. Dwell time determines how much cooling takes place prior to the deposition of the next layer. With relatively short dwell times, partial reheating cycles, in the so-called intrinsic in-process heat treatment, reduce the stress in the lower layers but build up more heat in the uppermost layers (Safyari et al., 2024).

Longer dwell times allow the lower regions to cool more uniformly, reducing residual stresses at the bottom, while the higher stresses are concentrated in the last layer. This is in agreement with observations from(Ahmad et al., 2020; Andrade Tatemoto et al., 2024), who found that dwell-time variation strongly affects stress distribution and microstructural evolution. While only one layer is simulated here, the model indeed shows thermal accumulation along the travel direction, as documented for multi-layer WAAM.

Thermal Gradients and Stress Development

All three variables have an impact on the magnitude of the thermal gradient-the principal cause of the development of stresses in WAAM. Increased heat input and lower speed increases the thermal gradients and tensile stresses. Longer cooling times flatten the gradients, while shorter dwells increase temperature build-up in higher layers. These results are in good agreement with classical welding mechanics (Lindgren, 2006), where constrained contraction upon cooling is responsible for tensile residual stresses.

These findings also resonate closely with established studies from (Andrade Tatemoto et al., 2024; Ding et al., 2015b; Kokare et al., 2024; Montevecchi et al., 2016b; Safyari et al., 2024), Confirming the reliability of the present thermo-mechanical simulation.

5 Conclusions

The thesis provided a numerical analysis of the thermal and mechanical behaviour of the Wire Arc Additive Manufacturing (WAAM) process with a single beam plate model. ANSYS Workbench was used to develop a coupled thermo-mechanical simulation with the temperature field generated during a thermal analysis as an input of the structural model. Goldak model of source of heat is a model that uses the Goldak double- ellipsoidal heat source model and these models were used to simulate the moving arc to predict heating and reheating during deposition in a realistic way.

It was demonstrated that the process formed a typical banana-shaped pool of melting material and temperature-affected region, where highest temperatures were more than 1500 degC at the centre of the weld. The high thermal gradients provided large expansion and contraction cycles and thus caused high residual stresses that were left up following cooling. The tensile stresses were concentrated on the weld bead, compressive stresses on the surrounding substrate were observed as was found in the work of previous researchers (Williams et al., 2016b). The parameter analysis showed that low heat input and prolonged dwell time served well in reducing residual stresses and enhancing bead stability as must have been the case with others. (Ding et al., 2015b), (Martina et al., 2012) and (Andrade Tatemoto et al., 2024).

Resting on the results, the following are the major contributions of this thesis:

The suitability of the Goldak model of double-ellipsoidal heat source in modelling thermal behaviour in WAAM of structural steels was validated.

- Created a simplified yet efficient coupled thermo-mechanical FEM model which can be used to forecast the temperature field and residual stress field in a single-bead deposition.
- Determined heat input, travel speed and dwell time as the most prominent variables that affect thermal gradients and residual stress development.

Can be used to offer practical ideas on decreasing distortion and enhancing process stability through tuning these important parameters.

To work in the future, the model can be developed in a variety of ways that are meaningful. It could be expanded to more complicated WAAM geometries (e.g. curved or variable-thickness components) in which thermal gradients might not behave in the same way. Other avenues are to consider the applicability of the model to other material classes other than structural steels such as stainless steels, nickel alloys, or aluminium alloys to extend the applicability of the framework. These extensions would augment the generalisation and usefulness of the modelling approach to industrial WAAM applications.

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